Multistrategy Region and Boundary Interaction Network for Salient Object Detection in Optical Remote Sensing Images

Yun Jia, *Member, IEEE*, Jie Zhao, Lin Ma, *Senior Member, IEEE*, and Lidan Yu

***Abstract*— Based on current theoretical insights and ideas in salient object detection in optical remote sensing images (RSI-SOD), the utilization of boundaries typically involves enhancing object features through boundary-guided assistance. However, this initial approach overlooks the intrinsic connections between regions and boundaries, which are essential for capturing rich contextual features and refining boundary details from a global perspective, thereby providing robust support for the mutual enhancement of information. To further capture the complex relationships between regions and boundaries, we propose a novel RSI-SOD network, the multistrategy region and boundary interaction network (MRBINet), which employs multiple strategies to analyze potential key information of the objects. Specifically, we design the boundary-region split (BRS) module to decompose image information and obtain features of regions and boundaries. Based on the obtained information, the boundary-guided reasoning refinement (BGRR) module and the multisource dynamic cross-attention (MDC) module are employed to explore intrinsic connections between regions and boundaries through graph reasoning and interactive attention, respectively, during the SOD process, effectively constraining diffuse regional information through robust boundary guidance. Finally, we utilize the collaborative enhancement cascade decoder (CECD) module to recombine and reinforce the refined regions and boundaries, aiming to restore the intrinsic features of the target object. Extensive experiments on three benchmark datasets demonstrate that our method outperforms classical approaches, showing competitive performance. Furthermore, the intrinsic features enhancement of region and boundary contributes to the performance improvement of the RSI-SOD method. The code and results for this work can be found at https://github.com/JieZzzoo/MRBINet**

***Index Terms*— Boundary, graph reasoning, optical remote sensing images (RSIs), region, salient object detection (SOD), Transformer.**

I. Introduction

SALIENT object detection (SOD), a crucial technique in computer vision tasks, aims to identify the most attention-grabbing objects in an image. Now, SOD has significantly advanced other downstream tasks, including but not limited to object detection [1], [2], object segmentation [3], image quality assessment [4], visual tracking [5], and medical segmentation [6].

In recent years, SOD has rapidly advanced in natural scenes. Methods for SOD in natural scene images (NSI-SOD) captured by handheld cameras and other media can be broadly classified into two categories: traditional handcrafted methods and deep learning methods enabled by large-scale annotated data. Traditional methods primarily rely on distinguishing objects based on color, texture, and orientation. Then, machine learning methods have also advanced these traditional techniques. However, despite a solid theoretical foundation, real-world scenes are diverse and multifaceted, with objects often embedded in noisy backgrounds, posing significant challenges for SOD. In contrast, deep learning methods based on large datasets can learn more comprehensive object information, effectively segregating salient objects from cluttered backgrounds and achieving satisfactory results. For instance, methods [7], [8] improved the U-shaped architecture by nesting and aggregating contextual information, significantly improving detection compared to traditional methods. Zhao et al. [9] proposed an edge-guided SOD framework that further models the intrinsic relationships within target regions, emphasizing the indispensable role of boundary information in accurately delineating salient objects. Nevertheless, the large computational load impedes further model development. However, these methods were originally designed for NSIs. While they can be adapted for optical remote sensing images (RSIs), their performance is not ideal, with errors in detail, as illustrated in Fig. 1. For example, the representative NSI-SOD method PA-KRN [10] fails to capture some crucial information when depicting the overall salient object. This indicates a significant difference between NSIs and optical RSIs, making SOD in optical RSIs (RSI-SOD) more challenging.

Based on the concept of NSI-SOD, several models focused on RSI-SOD have emerged, garnering significant attention. The semantic information of objects is progressively emphasized through the cascaded architecture, effectively enhancing the network's capability for subsequent information processing [11], [12]. However, these networks tend to overlook the critical role of boundary cues in the recognition process, placing exclusive emphasis on region-level feature extraction. Subsequently, several methods have focused on object morphology to further explore the constraining role of boundary

Received 7 November 2024; revised 29 April 2025; accepted 20 June 2025. Date of publication 14 July 2025; date of current version 22 July 2025. This work was supported by Yantai Science and Technology Innovation Development Plan under Grant 2022XDRH031. *(Corresponding author: Jie Zhao.)*

Yun Jia and Jie Zhao are with the School of Information and Electronic Engineering, Shandong Technology and Business University, Yantai 264005, China (e-mail: jiayun0223@sdtbu.edu.cn; jiezhao_99@163.com).

Lin Ma is with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150006, China (e-mail: malin@hit.edu.cn).

Lidan Yu is with the School of Computer and Information Engineering, Qilu Institute of Technology, Jinan 250200, China (e-mail: lidan_yu@163.com).

Digital Object Identifier 10.1109/TGRS.2025.3588415

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

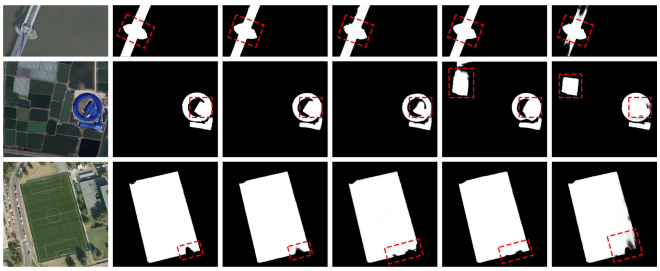


Fig. 1. Comparative analysis of our proposed method with classic RSI-SOD and NSI-SOD approaches. The red boxes highlight the failed salient information. (a) Optical RSI. (b) GT. (c) MRBINet (Ours). (d) ASTT [23]. (e) SFANet [22]. (f) PA-KRN [10].

information in mitigating semantic ambiguity during object representation. Previous works [13], [14] extracted region features from the encoded and designed a boundary extraction branch to capture information, providing implicit supervision to generate finer edge details. While early SOD methods in optical remote sensing predominantly relied on convolutional neural networks (CNNs), their intrinsic locality—due to the sliding window mechanism—limits the ability to capture long-range dependencies and elongated object structures, which are common in remote sensing imagery. To overcome these limitations, Transformer architectures have gained increasing attention in recent years for their powerful global modeling capabilities. By leveraging self-attention mechanisms, Transformers enable long-range contextual reasoning, making them particularly well-suited for modeling complex spatial relationships and multiscale structures in remote sensing scenes [17], [19]. In parallel, graph neural networks (GNNs) offer a flexible framework for encoding structural dependencies and topological relationships among image regions. Unlike traditional grid-based operations, GNNs excel at capturing high-order interactions, which are crucial for delineating salient objects with irregular shapes or indistinct boundaries, such as [20], [21], and [22]. However, existing models insufficiently exploit the intrinsic interplay between region and boundary features, and the topological structure of the enhanced representations formed through their integration remains underexplored. Fig. 1 illustrates ASTT [23] and SFANet [22], which are based on Transformer and graph, respectively. While both succeed in object localization, they fail to accurately segment object boundaries near adjacent regions due to external environmental influences.

To address the limitations of existing models, this work aims to achieve a more comprehensive integration and complementarity between regions and boundaries. We designed a multistrategy region and boundary interaction network (MRBINet). Specifically, we first separate the regions and boundaries from the optical RSIs, the boundary-region split (BRS) module. Then, we feed the regions and boundaries into the boundary-guided reasoning refinement (BGRR) module to further explore the topological information among images, maximizing the relationship between regions and boundaries. Subsequently, the boundaries and more structurally characterized regions are fed into the multisource dynamic cross-attention (MDC) module for cross-attention. This approach captures global features not solely from a single source, but by leveraging the complementary effects of both sources, significantly differing from existing Transformer-based RSI-SOD methods. Furthermore, we designed a collaborative enhancement cascade decoder (CECD) module that comprehensively integrates the extracted features, enhancing the representation of salient objects.

The main contributions of this work can be summarized as follows.

1) We propose an interactive RSI-SOD method named MRBINet, facilitating extensive interaction between regions and boundaries through a multistrategy processing approach. To the best of our knowledge, this is the first RSI-SOD approach that leverages Transformer- and graph-based cross-attention with regions and boundaries as the reference.
2) We introduce the BRS to extract region and boundary features from optical RSIs. Building on this foundation, the BGRR explores the complementary semantic relationships between regions and boundaries, effectively modeling topological structures. Additionally, the MDC captures intricate interactions between region interiors and boundaries, thus robustly guiding the delineation of diffuse regions with boundary information, resulting in more complete and coherent salient objects in the saliency maps (salmaps).
3) We propose the CECD to recombine the rich regions and boundaries obtained, enhancing information interaction in a multistage cascading manner to generate more discriminative features.

The remaining structure of this article is as follows. Section II describes related work, and Section III systematically explains our proposed framework and components, including the designed loss function. Section IV comprehensively reports the performance of our model and the necessity of all components, as well as failure cases. Finally, we conclude this article.

II. Related Works

A. SOD for RSIs

RSI-SOD has made significant contributions to the advancement of SOD. Due to the emergence of three classic datasets, ORSSD [11], EORSSD [12], and ORSI-4199 [24], numerous inspiring methods have been proposed successively, driving the development of haze removal in optical RSIs [25], SOD in hyperspectral images [26], and multimodal RSI salient detection [27].

Existing RSI-SOD mostly relies on CNN and occupies a dominant position. Dividing existing CNN-based RSI-SOD methods, they can be roughly categorized into two types: region-enhanced RSI-SOD and boundary-guided RSI-SOD. Some scholars attempt to enhance region features to strengthen salient object representation. Zhang et al. [12] advocate for the inclusion of pyramid attention modules in the encoder to progressively optimize features and attention clues in a cascading manner, achieving feature extraction and noise interference removal. AGNet [28] utilizes deep semantic information to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

assist shallow features, accurately describing the approximate location of salient objects, and strengthens detail expression through a self-refinement module. Shen et al. [29] advocate for assigning features with different receptive fields and feature description capabilities under different backgrounds through a multiscale strategy. Another class of methods considers the importance of boundary features, using it as an additional component to assist in better representing object information with region semantic information. For example, EMFINet [13] and MJRBM [24] advocate for integrating boundaries extracted from the encoder as a guiding approach with information encoded in each branch, effectively constraining the region features. Zhou et al. [14] design a parallel boundaries branch with the regions branch and use implicit supervision to encourage the network to learn more texture information. AESINet [30] adaptively cascades and interacts between boundaries extracted from shallow branches and deep semantic information, achieving efficient RSI-SOD. A plug-and-play boundary detection module to comprehensively capture the morphological changes of objects in optical RSIs from four perspectives, attempting to better complete missing boundary information for objects by Zhao et al. [31].

The aforementioned methods provide valuable insights into RSI-SOD. However, most of these ideas are built upon CNN, which, while satisfactory in capturing local features, inevitably suffers from shortcomings in obtaining global structural features. Therefore, it becomes crucial to have a framework based on both regions and boundaries, capable of capturing local features while also not missing out on global structural features.

B. Transformer for SOD

Transformer has gained immense popularity in natural language processing (NLP), prompting scholars to explore its application in the field of image processing. It addresses the limitations of CNN in acquiring information through progressive window views and effectively models long-distance relationships. Initially applied in NSI-SOD, the Transformer greatly alleviated barriers in acquiring global features. VST [32] constructed a pure Transformer framework for both RGB and RGB-D, executing regions and boundaries saliency predictions simultaneously through a multitask decoder. TF-SOD [33], utilizing local information as a basis, performs multiscale fusion and delivers it to Transformer to extract more global information. Ke Yun and Lin [34] build a salient detection network primarily using PVT as the encoder and refine the obtained features globally and locally, resulting in clearer object feature representation. Subsequently, Transformer's presence has been observed in RSI-SOD, achieving favorable results. HFANet [35] advocates for using CNN to extract shallow encoding information, followed by a Transformer for obtaining deep semantic information, thus mitigating interference from complex backgrounds. BMGT [36] utilizes Transformer as the primary encoder for acquiring global information and further supplementing local information. GeleNet [37] also employs a Transformer as the encoding structure but reconfigures the refined information, breaking the information barrier between channels and facilitating accelerated collaboration. TGDNet [18] advocates for extracting multiscale features through a global receptive field and refining them based on the characteristics of feature hierarchies. ASTT [23] designs an adaptive spatial tokenization Transformer encoder to extract global–local features, accurately sparsifying labels for each input image, and achieving competitive performance in RSI-SOD.

Although Transformer has garnered extensive attention in the RSI-SOD, existing work primarily focuses on extracting global information from region features, neglecting the auxiliary role of boundaries in the representation process. In response, we aim to incorporate both boundary and region features into the Transformer framework to obtain more precise global information, thoroughly explore the relationship between them, and ultimately enhance SOD.

C. Graph Reasoning for SOD

In recent years, both CNN and Transformer have made significant strides; however, their performance on non-Euclidean data structures has been less than satisfactory. The advent of GNNs has broken this impasse, introducing new possibilities. GCNs have been successfully applied to image processing, including downstream tasks such as image classification, detection, and segmentation. For instance, GCN was used as the backbone encoder to construct spatiotemporal graphs for further interpreting remote sensing change detection [45]. Multilevel superpixel-guided sparse GAT was designed to effectively utilize spatial topological structures for hyperspectral image classification by creating adjacency graphs at different stages by Li et al. [46]. GCN has also been widely introduced in medical segmentation to enhance feature extraction, as seen in works like [47] and [48].

In the field of RSI-SOD, GCN has also been employed. For example, Cong et al. [20] designed a relational reasoning network to enhance the localization of semantic relationships in both channel and spatial dimensions and utilized a multiscale attention module to restore detailed information. U2GL [21] incorporated GCN at the end of the encoder to discover topological relationships among objects from the encoded information, better-displaying object morphology. SFANet [22] used a trainable projection matrix to project feature maps onto graph representations, achieving deep semantic reasoning and modeling long-range dependencies and global context in the feature space to accurately identify salient objects. However, these models predominantly use region features as the primary data, neglecting the importance of boundary features in object feature representation.

To address this limitation, we aim to develop a more comprehensive graph reasoning module that fully exploits the synergy between boundaries and regions. This approach will enhance the representational capacity of salient object features in optical RSIs, enabling the restoration of more refined object characteristics.

III. Proposed Method

In this section, we provide a detailed description of the proposed MRBINet, the architecture of which is illustrated

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

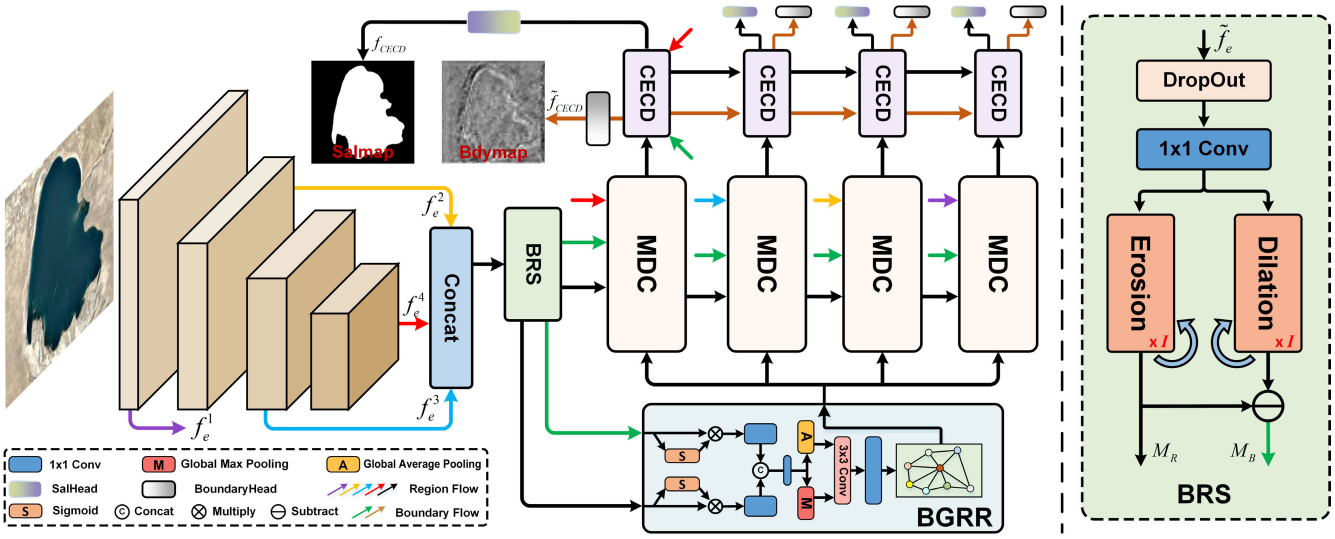


Fig. 2. Overview of the proposed MRBINet. The encoder employs Res2Net to extract both low- and high-level features from an optical RSI. Subsequently, the encoder features are fed into the BRS to extract boundary and region features. This feature is then processed through the BGRR and MDC for feature refinement. Next, the refined features are decoded by the CECD, yielding the final boundary and region prediction maps. Additionally, the multitask joint loss function implicitly constrains network randomness, continuously optimizing the framework's performance.

in Fig. 2. In Section III-A, we elaborate on the network architecture. In Sections III-B–III-E, we introduce the BRS module, BGRR module, MDC module, and CECD module. Following that, we explain the loss functions employed in this work.

A. Network Overview

As depicted in Fig. 2, MRBINet follows an encoder–decoder architecture, with the backbone network incorporating BRS, BGRR, MDC, and CECD. Given an optical RSI $I \in \mathbb{R}^{3\times256\times256}$, we utilize the Res2Net [49] encoder for initial feature extraction. To better adapt to the RSI-SOD, we discard the last adaptive average pooling layer and fully connected layer of Res2Net, dividing the remaining structure into four stages (following the design in [16] and [23]). This results in feature maps $f_e^i \in \mathbb{R}^{C_i\times H_i\times W_i}$, where i ranges from 1 to 4, $C_i = \{256, 512, 1024, 2048\}$, and H_i and W_i are both set to $256/2^{i+1}$. Objects in optical RSIs typically exhibit multiscale characteristics. To address this, we concatenate the encoded features f_e^i and apply 1×1 convolutions for feature smoothing within the concat module, effectively fusing multiple information to produce $\tilde{f}_e$. Next, we use the BRS, relying on morphological operations, to obtain boundary features M_{B} and region features M_{R}. Subsequently, M_{B} and M_{R} are processed by BGRR to obtain information with more topological features, resulting in f_{BGRR}. Then, f_{BGRR}, M_{B}, M_{R}, and f_e^i are fed into MDC to reinforce the feature representation of objects in an interactive attentional manner between boundaries and regions, enhancing the overall structure of the objects, to obtain f_{MDC}. Finally, we design the CECD to synergistically integrate the previously obtained boundary and region features. This is accomplished by exploiting their intrinsic relationship through a cascaded architecture, whereby each CECD generates predicted saliency maps (salmaps) f_{CECD} and boundary maps (bdymaps) $\tilde{f}_{\mathrm{CECD}}$. Additionally, to facilitate the network in learning more precise object features, we design a multitask joint supervision loss function, implicitly constraining the randomness in the learning process of this framework.

B. BRS Module

In optical RSIs, regional features are crucial for accurately localizing objects, while boundary features play a key role in precisely delineating object extents. Under the effective guidance of boundary features, the variability of regional representations can be suppressed, making them more consistent with real-world structures. Inspired by [50, Ch. 9.2], we adopt a boundary-region separation strategy to decompose prominent objects in optical RSIs. According to the morphological image processing framework, erosion operations can suppress ambiguous boundary pixels and progressively refine object contours through iterative shrinking, thus enhancing the compactness of core regions. In contrast, dilation operations expand object boundaries, fill discontinuities (e.g., holes or fractures), and reconnect fragmented regions to strengthen spatial continuity. By subtracting the eroded result from the dilated result, accurate boundary information can be effectively extracted. In this work, we embed morphological principles into a deep network architecture through the proposed BRS, where multilevel fused feature maps undergo iterative boundary-region interaction guided by morphological operations. Prior to this, similar to [51], we first concatenate the features f_e^2, f_e^3, and f_e^4 obtained by the encoder and perform a 1×1 convolution to smooth the features, facilitating effective interaction between different channels of features, resulting in $\tilde{f}_e$. This mechanism enables hierarchical encoding of multiscale contextual features, which provides discriminative representations to empower subsequent BRS with enhanced localization accuracy for both core regions and boundary delineation of target areas while inherently suppressing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

noise interference. Next, the processed features are fed into the BRS, which dynamically balances multiscale feature contributions to decouple object boundaries and internal regions. Then, we perform a 1×1 convolution for feature smoothing, obtaining $\tilde{f}'_e$. Subsequently, we perform binarization on $\tilde{f}'_e$ to maximize the efficacy of morphological operations on salient objects. Erosion and dilation operators with a 3×3 kernel matrix of all ones are then applied to separate regions and boundaries, yielding M_{R} and M_{B}, respectively. Fundamentally, dilation expands the regions while erosion contracts them, with the difference between these operations defining the boundary regions. In this process, we set I separation iterations, with each iteration expanding or eroding the mask edge by one pixel. It is the continuous refinement of these pixels that enables the boundaries and regions of the object to be more accurate. The separation process described above can be represented as

$$\tilde{f}_e = \mathrm{conv}_{1\times1}\left(\varpi\left(f_e^2, f_e^3, f_e^4\right)\right) \tag{1}$$

$$\tilde{f}'_e = \mathrm{conv}_{1\times1}\left(\wp\left(\tilde{f}_e\right)\right) \tag{2}$$

$$M_{\mathrm{R}} = E\left(\tilde{f}'_e\right) \times I \tag{3}$$

$$M_{\mathrm{B}} = \left(D\left(\tilde{f}'_e\right) \times I - M_{\mathrm{R}}\right) \tag{4}$$

where ϖ represents the concatenation, $\mathrm{conv}_{1\times1}$ denotes the 1×1 convolution, and $\wp$ indicates the dropout. E and D represent the erosion operator and dilation operator, respectively, and both operators are applied the same number of times, which is I; here, we set it to 3.

C. BGRR Module

The reasoning network leverages advanced topological processing, enabling broad applicability. While internal region information captures object attributes, intrinsic region-boundary connections further enhance feature representation. Boundary information constrains region diffusion, while regions fill boundary interiors—an aspect often overlooked. To address this, we designed the BGRR to capture intricate pixel correlations, particularly for complex topologies. First, we parallelly utilize the sigmoid gating mechanism to spatially suppress local interference (e.g., boundary ambiguity and regional misclassification) while attenuating noise introduced during BRS-driven feature decoupling. This feature refinement synergistically enhances spatial discriminability, yielding optimized M_{BS} and M_{RS} representations with preserved geometric fidelity. Subsequently, the feature maps undergo channel-wise recalibration through 1×1 convolution, which establishes optimal linear combinations of multiscale contextual features to enhance pixel-wise discriminative power while preserving spatial coherence. Finally, the obtained information is concatenated and again passed through a 1×1 convolution to obtain M_{BRG}. This process can be represented as

$$M_{\mathrm{RS}} = \mathrm{sigmoid}\left(M_{\mathrm{R}}\right) \odot M_{\mathrm{R}} \tag{5}$$

$$M_{\mathrm{BS}} = \mathrm{sigmoid}\left(M_{\mathrm{B}}\right) \odot M_{\mathrm{B}} \tag{6}$$

$$M_{\mathrm{BRG}} = \mathrm{conv}_{1\times1}(\varpi(conv_{1\times1}(M_{\mathrm{RS}}), \mathrm{conv}_{1\times1}(M_{\mathrm{BS}}))) \tag{7}$$

where the sigmoid represents the sigmoid operation and $\odot$ denotes the element-wise multiplication.

To preserve the overall distribution of features and emphasize the salient parts, we utilize global average pooling and global max pooling to process M_{BRG} and better integrate multiscale features, enhancing the richness of representation. Subsequently, we employ 3×3 convolutional kernels followed by 1×1 convolutions to further extract refined features from the fused output, enabling the model to learn complex feature patterns more comprehensively, obtaining $\hat{M}_{\mathrm{BRG}}$. This process can be described as

$$\hat{M}_{\mathrm{BRG}} = \mathrm{conv}_{1\times1}(\mathrm{conv}_{3\times3}(M(M_{\mathrm{BRG}}) + A(M_{\mathrm{BRG}})) \tag{8}$$

where M represents global max pooling, A represents global average pooling, and $\mathrm{conv}_{3\times3}$ represents the 3×3 convolution. Through these operations, pixel features become more persuasive in expressing both regions and boundaries. Naturally, the connectivity between the object boundary and region features is strengthened. Then, we map pixels with similar features in $\hat{M}_{\mathrm{BRG}}$ to vertices in a graph L. Subsequently, each vertex is aggregated based on the relevance of information and forms an adjacency matrix P. Then, we perform graph reasoning on P and $\hat{M}_{\mathrm{BRG}}$ using GCN, which explicitly models relational dependencies between vertices through spectral-domain filtering. Specifically, the GCN propagates information across nodes by aggregating neighborhood features weighted by their relevance scores in P, thereby capturing long-range contextual interactions that are critical for resolving ambiguous boundaries. This is followed by the activation function ReLU to introduce nonlinearity and enhance discriminative feature learning. Finally, the obtained features are projected back to restore the original feature form. By propagating information between vertices, higher level semantic information can be learned to further infer potential relationships within the image. This process can be represented as

$$f_{\mathrm{BGRR}} = O_{\mathrm{Rep}}\left(\sigma\left(O_{\mathrm{GCN}}\left(P, \hat{M}_{\mathrm{BRG}}, \hat{w}\right)\right)\right) \tag{9}$$

where $\hat{w}$ represents the trainable weights of GCN, $\hat{M}_{\mathrm{BRG}}$ reflects the features input to BGRR prior to processing, P is the adjacency matrix, O_{GCN} denotes GCN reasoning, σ stands for ReLU, and O_{Rep} signifies the reprojection operation. The resulting feature f_{BGRR} combines global and local features, and through GCN, it further captures complex relationships between features, enhancing the accuracy and robustness of SOD.

D. MDC Module

The boundaries and regions represent distinct features, and relying solely on one of these sources in a Transformer can lead to omissions. To address this, we utilize cross-attention to fully leverage the precise localization from boundary features and the contextual information from the regions, as illustrated in Fig. 3. This fusion approach maintains boundary accuracy while enhancing semantic understanding of the regions, thereby improving detection accuracy and robustness. First, the previously extracted features are further integrated to enrich both boundaries and regions. For boundary features, we align the features M_{B} extracted by BRS with the encoder's output features and multiply them to reinforce the boundary components within the origin features, resulting in M'_{B}. For

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

region features, we combine the regions output from BRS or the previous MDC with f_{BGRR} to obtain M'_{R}, capturing object characteristics at different levels and enhancing detection performance. Subsequently, M'_{B} and M'_{R} are simultaneously fed into the spatial-frequency enhanced (SFE) module to reduce noise interference. The SFE consists of two branches. The first branch incorporates the fast Fourier transform (FFT) and inverse fast Fourier transform (iFFT). Specifically, image signals are transformed from the spatial domain to the frequency domain via FFT, where high-frequency random noise is separated through spectral analysis. A learnable frequency-domain feature selection mechanism retains effective frequency band information. The processed critical features are then reconstructed into spatial-domain signals through iFFT, providing enhanced representations for subsequent multiscale feature fusion. The second branch utilizes depthwise separable convolution (DWConv) to process spatial-domain features, effectively capturing details and boundary features. Finally, the features from both branches are integrated, combining global frequency-domain features with local spatial features, enhancing feature representation. The outputs from this process are $f^{\mathrm{B}}_{\mathrm{SFE}}$ and $f^{\mathrm{R}}_{\mathrm{SFE}}$. This process can be described as

$$f^{\mathrm{B}}_{\mathrm{SFE}} = \mathrm{iFFT}\big(\mathrm{FFT}\big(M'_{\mathrm{B}}\big)\cdot w\big) + \lambda\big(M'_{\mathrm{B}}\big) \tag{10}$$

$$f^{\mathrm{R}}_{\mathrm{SFE}} = \mathrm{iFFT}\big(\mathrm{FFT}\big(M'_{\mathrm{R}}\big)\cdot w\big) + \lambda\big(M'_{\mathrm{R}}\big) \tag{11}$$

where FFT represents the fast Fourier transform, iFFT denotes the inverse fast Fourier transform, w represents the learnable weights in the frequency-domain feature selection mechanism, and λ denotes the DWConv. Next, $f^{\mathrm{B}}_{\mathrm{SFE}}$ and $f^{\mathrm{R}}_{\mathrm{SFE}}$ underwent LayerNorm, followed by processing through three parallel branches. The first branch includes a 1×3 convolution followed by a 3×1 convolution. The second branch comprises a 1×5 convolution followed by a 5×1 convolution. The third branch consists of a 1×7 convolution followed by a 7×1 convolution. The features from these three branches are then concatenated and processed with a 1×1 convolution to adjust the channel number, enhancing the correlation at the same spatial positions across channels, resulting in f_{B} and f_{R}. Following this, f_{B} and f_{R} are processed in the ViT [52] to obtain two sets of matrices (V_1, K_1, and Q_1) and (V_2, K_2, and Q_2). Unlike existing methods, we employ a cross-attention mechanism to establish connections between boundary and region features. By querying the corresponding objects with their own key values, attention is computed to effectively combine the features, reinforcing the feature relationship network, resulting in f_{CAB} and f_{CAR}. Subsequently, a 3×3 convolution is used to enhance and combine the features. This process can be described as

$$\mathrm{Attn}(K, Q, V) = \mathrm{Softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V \tag{12}$$

$$f_{\mathrm{CAB}} = \mathrm{Attn}(K_1, Q_2, V_1) \tag{13}$$

$$f_{\mathrm{CAR}} = \mathrm{Attn}(K_2, Q_1, V_2) \tag{14}$$

$$f_{\mathrm{MDC}} = \varpi\left(\mathrm{conv}_{3\times3}(f_{\mathrm{CAB}}), \mathrm{conv}_{3\times3}(f_{\mathrm{CAR}})\right) \tag{15}$$

where Softmax denotes the softmax operation and f_{MDC} is the out of MDC.

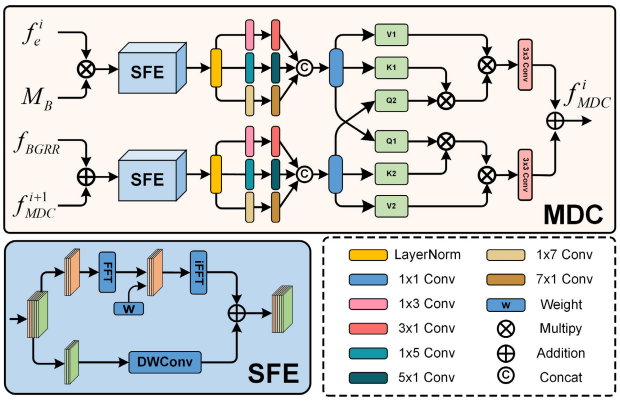


Fig. 3. Illustration of MDC and SFE.

E. CECD Moudle

The previous modules have already integrated the intrinsic connections between boundary and region features using various strategies. However, during long-range feature transmission and processing, feature loss and noise introduction are inevitable. To mitigate these adverse effects, we attempt to use multisource feature aggregation cooperatively to lay the groundwork for the final detection of salient objects, as illustrated in Fig. 4. First, directional convolutional kernels are employed to simultaneously extract features from four directions—horizontal, vertical, 45°, and 135°, and the features are concatenated for integration, better matching the complexity of optical RSIs. Subsequently, different strategies are adopted for boundary and region features. Boundary features $\tilde{f}^{i+1}_{\mathrm{CECD}}$, which contain less data, are processed with a sigmoid and fused with previous features to obtain $\tilde{f}^{i}_{\mathrm{CECD}}$. For region features $\tilde{f}^{i+1}_{\mathrm{CECD}}$, channel attention (CA) and spatial attention (SA) [53] are applied to obtain f_{RCS}. Then, $\tilde{f}^{i}_{\mathrm{CECD}}$ and f_{RCS} are concatenated and combined using a 3×3 convolution, resulting in f_{BRE}. To enhance semantic information, the corresponding MDC output f_{MDC} is processed with 3×3 and 1×1 convolutions to obtain f'_{MDC}. Next, the relationship between f'_{MDC} as Q and f_{BRE} as K and V, processed in the ViT [52], is explored to study pixel correlations, resulting in $\vec{f}_{\mathrm{CECD}}$. The 1×1 convolutional layers enable cross-channel feature recalibration, thereby generating discriminative representations denoted as f_{CECD}. This process can be described as follows:

$$\vec{f}_{\mathrm{CECD}} = \mathrm{Attn}(K, Q, V) \tag{16}$$

$$f_{\mathrm{CECD}} = \mathrm{conv}_{1\times1} \cdot \vec{f}_{\mathrm{FECD}} \tag{17}$$

where Attn represents the multihead self-attention mechanism in the ViT and f_{CECD} refers to the saliency-aware feature map generated by the CECD.

F. Multitask Joint Supervised Loss Function

We designed a multitask joint supervised loss function tailored to the framework proposed in this work, imposing privacy supervision constraints on both boundary and region features. Additionally, we introduced a weight compensation mechanism to optimize these two tasks. For region features, we employed binary cross-entropy (BCE) loss, intersection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

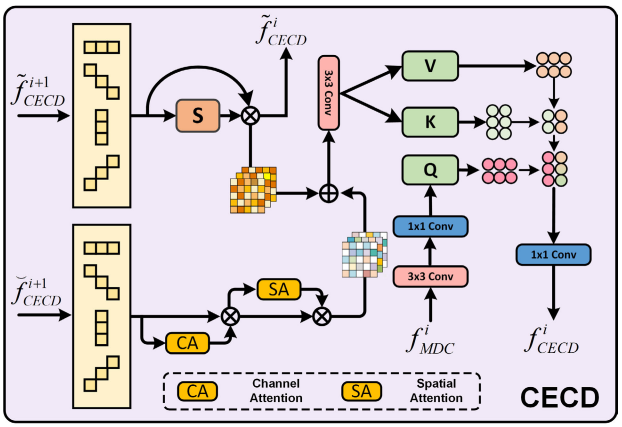


Fig. 4. Illustration of CECD.

over union (IoU) loss, and focal loss (Floss) for calculation. The mathematical expression for this combination is

$$\mathcal{L}_{\text{region}} = \sum_{i=1}^{4} \left(\mathcal{L}^i_{\text{BCE}}(S, \text{GT}) + \gamma \cdot \mathcal{L}^i_{\text{IoU}}(S, \text{GT}) + \psi \cdot \mathcal{L}^i_{\text{Floss}}(S, \text{GT}) \right) \tag{18}$$

where S denotes the predicted saliency maps, GT represents the ground truth saliency maps, $\mathcal{L}_{\text{BCE}}$ is the BCE loss, $\mathcal{L}_{\text{IoU}}$ is the IoU loss, and $\mathcal{L}_{\text{Floss}}$ is the focal loss. Here, γ and ψ correspond to the contribution weights of $\mathcal{L}_{\text{IoU}}$ and $\mathcal{L}_{\text{Floss}}$ in the overall network loss, respectively. $\mathcal{L}_{\text{region}}$ is the overall loss for region features. For boundary features, considering that boundary prediction is a relatively simple binary classification problem, we only use BCE to supervise it, resulting in the overall loss $\mathcal{L}_{\text{boundary}}$ for boundary features. The mathematical expression designed for this is

$$\mathcal{L}_{\text{boundary}} = \sum_{i=1}^{4} \left(\mathcal{L}^i_{\text{BCE}}(B, \text{BT}) \right) \tag{19}$$

where B represents the predicted boundary maps (bdymaps), BT represents the boundary truth maps, and $\mathcal{L}_{\text{boundary}}$ denotes the overall loss for boundary features. In summary, the multi-task joint supervised loss function proposed in this work is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{region}} + \mathcal{L}_{\text{boundary}} \tag{20}$$

where $\mathcal{L}_{\text{total}}$ represents the overall loss of the network. Finally, Algorithm 1 elaborates on the inference details of the proposed framework in this work.

Algorithm 1 Training Framework for MRBINet

Input: optical RSI $I \in \mathbb{R}^{3\times256\times256}$
Output: Salmap $S \in \mathbb{R}^{1\times256\times256}$, Bdymap $B \in \mathbb{R}^{1\times256\times256}$

while n in epoch N **do**
  Step1: Backbone extracts primary encoding features f_e^i, i = 1 to 4
  Step2: BRS iteratively extracts boundary and region features from the optical RSI through I=3 internal iterations
  $\tilde{f}'_e \Leftarrow conv_{1\times1}(\wp(conv_{1\times1}(\varpi(f_e^2, f_e^3, f_e^4))))$
  $M_B \Leftarrow (D(\tilde{f}'_e)\times I - M_R), M_R \Leftarrow E(\tilde{f}'_e)\times I$
  Step3: BGRR models the relationship between boundary and region features
  $f_{BGRR} \Leftarrow BGRR(M_R, M_B)$
  Step4: MDC performs cross-attention between refined region features and enhanced boundary features.
  if i =4 **then**
    $f^i_{MDC} \Leftarrow MDC(f^i_e, M_B, f_{BGRR}, M_R)$
  end if
  for i in 3,2,1 **do**
    $f^i_{MDC} \Leftarrow MDC(f^i_e, M_B, f_{BGRR}, f^{i+1}_{MDC})$
  end for
  Step5: CECD integrates the strengthened boundary and region features in a multi-level cascade
  if i = 4 **then**
    $f^i_{CECD}, \tilde{f}^i_{CECD} \Leftarrow CECD(M_B, f^i_e, f^i_{MDC})$
  end if
  for i in 3,2,1 **do**
    $f^i_{CECD}, \tilde{f}^i_{CECD} \Leftarrow CECD(\tilde{f}^{i+1}_{CECD}, \check{f}^{i+1}_{CECD}, f^i_{MDC})$
  end for
  Return S and B
  Compute the network loss to continually optimize the performance of MRBINet
end while

IV. Experimental Analysis

A. Datasets and Implementation Details

1) Datasets: To evaluate the effectiveness and complexity of our proposed model, we conducted a comprehensive analysis on three RSI-SOD datasets: ORSSD [11], EORSSD [12], and ORSI-4199 [24]. ORSSD is the first publicly available dataset in the RSI-SOD and has gained widespread adoption and application. It comprises 800 optical RSIs, with 600 images used for training and the remaining 200 images used for testing model performance. These images cover urban, forest, coastline, and terrain scenes. Subsequently, to enrich the variety and quantity of optical remote sensing datasets, EORSSD was introduced as an extension of ORSSD, expanding from 800 to 2000 images. These images were divided into 1400 training samples and 600 testing samples, featuring additional scenes such as rural areas, rivers, and mountains. Following this, the introduction of ORSI-4199 addressed the shortcomings of optical remote sensing datasets in terms of quantity and scene variety. It comprises 4199 images, making it one of the largest datasets to date. This dataset includes 2000 training samples and 2199 testing samples. Notably, the creators of these datasets annotated the salient objects in the images from an expert perspective, greatly facilitating the research process for subsequent researchers.

2) Evaluation Criteria: As shown in Tables I and II and Fig. 5, we provide a comprehensive quantitative comparison of our proposed model against state of the art (SOTA) across three datasets. Specifically, we employ extensive classical evaluation metrics, such as S-measure (S_α, $\alpha = 0.5$),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE RESULTS ON EORSSD [12] AND ORSSD [11]. THERE ARE 40 SOTA METHODS THAT CAN BE CLASSIFIED INTO THREE TRADITIONAL NSI-SOD (T.N.), 13 CNN-BASED NSI-SOD (C.N.), THREE TRADITIONAL RSI-SOD (T.R.), 16 CNN-BASED RSI-SOD (C.R.), THREE GRAPH-BASED RSI-SOD (G.R.), AND TWO TRANSFORMER-BASED RSI-SOD (Tr.R.). OUR METHOD DENOTES THE TRANSFORMER-BASED, GRAPH-BASED RSI-SOD (TrG.R.). ↑/↓ MEANS THAT A LARGER/SMALLER SCORE IS BETTER. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Methods	Publication	Type	Speed (*fps*) ↑	#Params (M) ↓	FLOPs (G) ↓	EORSSD [12]								ORSSD [11]							
						$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
RRWR [55]	2015 CVPR	T.N.	0.3	-	-	.5992	.3993	.3686	.3344	.6894	.5943	.5639	.1677	.6835	.5590	.5125	.4874	.7649	.7017	.6949	.1324
DSG [56]	2017 TIP	T.N.	0.6	-	-	.6428	.5259	.4597	.4012	.7286	.6954	.6269	.1246	.7195	.6238	.5747	.5657	.7912	.7337	.7538	.1024
VST [32]	2021 ICCV	T.N.	23.0	44.0	23.2	.9203	.8724	.8257	.7015	.9736	.9417	.8872	.0067	.9318	.8999	.8710	.8126	.9755	.9596	.9417	.0100
DSS [57]	2017 CVPR	C.N.	8.0	62.2	114.6	.7874	.6868	.5801	.4597	.9205	.7631	.6933	.0186	.8262	.7467	.6962	.6206	.8860	.8362	.8085	.0363
RADF [58]	2018 AAAI	C.N.	7.0	62.5	214.2	.8188	.7473	.6582	.4933	.9162	.8567	.7160	.0168	.8259	.7619	.6856	.5730	.9130	.8298	.7678	.0382
EGNet [9]	2019 ICCV	C.N.	9.0	108.1	159.0	.8601	.8453	.6967	.5379	.9570	.8775	.7566	.0110	.8774	.8492	.7500	.7415	.9731	.9013	.8851	.0216
PoolNet [59]	2019 CVPR	C.N.	25.0	53.6	49.0	.8869	.8347	.7905	.6723	.9524	.9167	.8647	.0102	.9026	.8687	.8433	.7861	.9509	.9341	.9205	.0168
CPDNet [60]	2019 CVPR	C.N.	84.0	47.9	9.4	.8874	.8093	.7660	.6638	.9390	.8977	.8663	.0111	.8954	.8524	.8243	.7717	.9438	.9207	.9211	.0187
MINet [8]	2020 CVPR	C.N.	12.0	47.6	146.3	.9040	.8344	.8174	.7705	.9442	.9346	.9243	.0093	.9040	.8761	.8574	.8251	.9545	.9454	.9423	.0144
GCPA [61]	2020 AAAI	C.N.	23.0	67.1	54.3	.8869	.8347	.7905	.6723	.9524	.9167	.8647	.0102	.9026	.8687	.8433	.7861	.9509	.9341	.9205	.0168
GateNet [62]	2020 ECCV	C.N.	25.0	100.0	108.3	.9114	.8566	.8228	.7109	.9610	.9385	.8909	.0095	.9186	.8913	.8679	.8229	.9719	.9562	.9428	.0131
U2Net [7]	2020 PR	C.N.	25.0	44.0	37.0	.9199	.8732	.8329	.7221	.9649	.9373	.8989	.0076	.9162	.8738	.8492	.8038	.9539	.9387	.9326	.0166
F3Net [63]	2020 AAAI	C.N.	-	25.0	8.0	.9020	.8358	.7924	.7330	.9573	.9421	.9153	.0089	.9170	.8762	.8526	.8164	.9651	.9559	.9441	.0152
SUCA [64]	2020 TMM	C.N.	24.0	117.7	56.4	.8998	.8229	.7949	.7260	.9520	.9277	.9082	.0097	.8989	.8484	.8237	.7748	.9584	.9400	.9194	.0145
PA-KRN [10]	2021 AAAI	C.N.	16.0	141.1	617.7	.9192	.8639	.8358	.7993	.9619	.9536	.9416	.0104	.9239	.8890	.8727	.8548	.9680	.9620	.9579	.0139
PGNet [65]	2022 CVPR	C.N.	43.0	73.7	77.4	.8757	.7887	.7775	.7402	.9315	.9178	.8858	.0153	.8894	.8338	.8192	.7921	.9420	.9218	.9105	.0225
VOS [66]	2018 GRSL	T.R.	-	-	-	.5082	.2765	.2107	.1836	.5982	.4886	.4767	.2096	.5366	.3471	.2717	.2633	.6514	.5352	.5826	.2151
CMC [67]	2019 RS	T.R.	-	-	-	.5798	.3268	.2692	.2007	.6803	.5894	.4890	.1057	.6033	.3913	.3454	.3108	.7064	.6417	.5996	.1267
SMFF [68]	2019 IJRS	T.R.	-	-	-	.5401	.5176	.2992	.2083	.7744	.5197	.5014	.1434	.5312	.4417	.2684	.2496	.7402	.4920	.5676	.1854
LVNet [11]	2019 TGRS	C.R.	1.4	207.0	-	.8644	.8051	.7328	.6306	.9279	.8801	.8465	.0145	.8815	.8414	.7995	.7506	.9456	.9259	.9195	.0207
SARNet [69]	2021 RS	C.R.	47.0	25.9	129.7	.9240	.8719	.8541	.8304	.9620	.9555	.9536	.0099	.9134	.8850	.8619	.8512	.9557	.9477	.9464	.0187
DAFNet [12]	2022 TIP	C.R.	26.0	29.4	68.5	.9166	.8614	.7845	.6427	**.9861**	.9291	.8446	.0060	.9191	.8928	.8511	.7876	.9771	.9539	.9360	.0113
EMFINet [13]	2022 TGRS	C.R.	25.0	107.3	480.9	.9290	.8720	.8486	.7984	.9711	.9604	.9501	.0084	.9366	.9002	.8856	.8617	.9737	.9671	.9663	.0109
ERPNet [14]	2022 TCYB	C.R.	50.0	56.5	87.2	.9210	.8632	.8304	.7554	.9603	.9401	.9228	.0089	.9254	.8974	.8745	.8356	.9710	.9566	.9520	.0135
MSCNet [70]	2022 ICPR	C.R.	55.0	3.3	23.4	.9086	.8539	.8151	.7553	.9689	.9551	.9329	.0090	.9227	.9107	.8676	.8350	.9754	.9653	.9584	.0129
AGNet [28]	2022 ICANN	C.R.	-	26.6	9.7	.9287	.8785	.8516	.8021	.9752	.9656	.9540	.0067	.9389	.9098	.8956	.8723	.9811	.9728	.9729	.0091
FSMINet [29]	2022 GRSL	C.R.	17.7	3.6	9.7	.9255	.8678	.8436	.8015	.9666	.9567	.9490	.0079	.9361	.9041	.8878	.8710	.9759	.9672	.9693	.0101
HFANet [35]	2022 TGRS	C.R.	26.0	60.5	68.3	**.9380**	**.8876**	.8681	.8365	.9740	.9679	.9644	.0070	.9399	.9112	.8981	.8819	.9770	.9712	.9722	.0092
MJRBM [24]	2022 TGRS	C.R.	32.0	43.5	95.7	.9197	.8701	.8239	.7066	.9646	.9350	.8897	.0099	.9204	.9025	.8566	.8022	.9623	.9415	.9328	.0145
ACCoNet [54]	2023 TCYB	C.R.	81.0	102.6	368.6	.9290	.8837	.8552	.8008	.9727	.9653	.9454	.0074	.9437	.9149	.8971	.8806	.9796	.9754	.9721	.0088
SeaNet [16]	2023 TGRS	C.R.	96.0	2.8	1.7	.9208	.8649	.8519	.8304	.9710	.9651	.9602	.0073	.9260	.8942	.8772	.8625	.9767	.9722	.9670	.0105
CorrNet [15]	2023 TGRS	C.R.	100.0	4.1	21.1	.9289	.8778	.8620	.8311	.9696	.9646	.9593	.0083	.9380	.9129	.9002	.8875	.9790	.9746	.9721	.0098
PCDNet [71]	2023 TGRS	C.R.	30.2	17.7	52.9	.9346	.8830	.8590	.8088	.9744	.9645	.9507	.0069	.9444	.9138	.9014	.8839	.9809	.9742	.9731	.0094
ADSTNet [31]	2024 JSTARS	C.R.	39.5	62.1	27.7	.9311	.8804	**.8716**	**.8532**	**.9769**	.9709	**.9681**	.0065	.9379	.9124	**.9042**	**.8979**	.9807	.9740	**.9785**	.0086
TSCNet [72]	2024 TCAS-II	C.R.	-	-	-	**.9383**	**.8900**	**.8740**	**.8509**	.9765	**.9717**	.9688	.0061	.9428	**.9198**	.9030	.8879	**.9851**	**.9804**	.9760	**.0081**
RRNet [20]	2022 TGRS	G.R.	41.5	68.0	190.0	.9266	.8781	.8351	.7251	.9716	.9415	.9034	.0075	.9260	.8923	.8667	.8162	.9687	.9498	.9434	.0114
AESINet [30]	2023 TGRS	G.R.	23.0	-	-	.9347	.8792	.8496	.7909	.9757	.9647	.9480	.0064	**.9455**	.9160	.8962	.8640	**.9814**	.9741	.9689	.0085
SFANet [22]	2024 TGRS	G.R.	-	25.1	7.7	.9349	.8833	.8680	.8492	**.9769**	**.9726**	.9669	**.0058**	**.9453**	**.9192**	**.9063**	**.8984**	**.9830**	**.9789**	**.9765**	**.0077**
ASTT [23]	2023 TGRS	Tr.R.	13.0	44.6	69.5	.9253	.8741	.8297	.7534	.9757	.9584	.9247	**.0059**	.9348	.9061	.8811	.8456	.9795	.9699	.9587	.0094
BMGT [36]	2023 IJRS	Tr.R.	35.0	3.8	22.7	.9292	.8829	.8409	.7245	.9740	.9484	.8972	**.0058**	.9368	.9105	.8871	.8370	.9781	.9610	.9529	.0092
Ours	-	TrG.R.	9.0	32.4	42.8	**.9351**	**.8852**	**.8768**	**.8626**	**.9766**	.9721	**.9729**	**.0056**	**.9474**	**.9199**	**.9110**	**.9058**	**.9851**	**.9812**	**.9793**	**.0069**

TABLE II

18 REPRESENTATIVE MODELS FOR FURTHER EVALUATION ON THE ORSI-4199 [24]. ↑/↓ MEANS THAT A LARGER/SMALLER SCORE IS BETTER. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE

Methods	ORSI-4199 [24]							
	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{mean}\uparrow$	$E_\xi^{adp}\uparrow$	$\mathcal{M}\downarrow$
VST [32]	.8790	.8717	.8524	.7947	.9481	.9348	.8997	**.0281**
EGNet [9]	.8464	.8267	.8041	.7650	.9161	.8947	.8620	.0440
PoolNet [59]	.8271	.8010	.7779	.7382	.8964	.8676	.8531	.0541
GateNet [62]	.8680	.8626	.8414	.7946	.9369	.9199	.8816	.0357
SUCA [64]	.8794	.8692	.8590	.8415	.9438	.9356	.9186	.0304
PA-KRN [10]	.8491	.8415	.8324	.8200	.9280	.9168	.9063	.0382
DAFNet [12]	.8552	.8458	.8261	.7819	.9220	.9007	.8905	.0396
EMFINet [13]	.8675	.8584	.8479	.8186	.9340	.9257	.9136	.0330
MJRBM [24]	.8593	.8493	.8309	.7995	.9311	.9102	.8891	.0374
ERPNet [14]	.8670	.8553	.8374	.8024	.9290	.9149	.9024	.0357
CorrNet [15]	.8623	.8560	.8513	.8534	.9330	.9206	.9142	.0366
ACCoNet [54]	**.8805**	.8688	.8547	.8372	.9424	.9352	.9109	.0320
ASTT [23]	**.8828**	**.8788**	**.8673**	.8527	**.9512**	**.9435**	**.9216**	**.0273**
SeaNet [16]	.8722	.8653	.8591	.8556	.9426	.9363	.9197	.0308
AESINet [30]	.8755	.8726	.8626	.8246	.9459	.9372	.9193	.0305
ADSTNet [31]	.8710	.8698	.8653	**.8655**	.9433	.9356	.9212	.0318
TSCNet [72]	.8783	**.8771**	**.8703**	.8634	**.9486**	**.9418**	**.9243**	.0295
SFANet [22]	.8761	.8710	.8659	**.8647**	.9447	.9385	.9202	.0292
Ours	**.8824**	**.8800**	**.8755**	**.8755**	**.9489**	**.9425**	**.9220**	**.0268**

mean absolute error (MAE, $\mathcal{M}$), F-measure (F_β and $\beta^2 = 0.3$), and E-measure (E_ξ). S-measure evaluates the structural similarity between the generated predicted saliency map (S) and the ground truth (GT) map, focusing on both object-aware and region-aware structures. MAE calculates the average absolute difference between the S and GT, providing an indication of the overall discrepancy between the generated saliency map and the true labels. F-measure combines precision and recall into a single metric to evaluate the accuracy of the saliency detection. E-measure, a relatively new metric for evaluating matching performance, compares the similarity between the S and the GT while considering both local and global levels of saliency detection results. To visually demonstrate the strengths and weaknesses of our model, we also use precision–recall (P–R) curves and F-measure curves for qualitative analysis. Specifically, the P–R curve closer to (1, 1) and an F-measure curve approaching the y-axis value of 1 indicate superior model performance.

3) Implementation Details: The proposed MRBINet was trained and tested using the PyTorch framework on a Linux platform with an Intel[1] Xeon[1] Silver 4210 CPU and an NVIDIA V100 GPU. To reduce training time and achieve higher performance, we initialized the backbone of this framework with pretrained weights of Res2Net trained on ImageNet. Each image was resized to a uniform size of 256 × 256. We employed the Adam optimizer to update model parameters, with a batch size and epoch defined as 8 and 50, respectively. The learning rate was set to $1e^{-4}$, and to facilitate model convergence to optimal values, we decreased the learning rate by a factor of 10 after 30 epochs, reducing it to $1e^{-5}$. Furthermore, similar to ACCoNet [54], we utilized data augmentation techniques to further augment the dataset. The sets of ORSSD, EORSSD, and ORSI-4199 were augmented to 4800, 11 200, and 16 000 images, respectively.

B. Comparison With SOTA Methods

To comprehensively and fairly evaluate the proposed model, we compared it with 40 SOTA methods across multiple dimensions. These include three traditional NSI-SOD methods

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

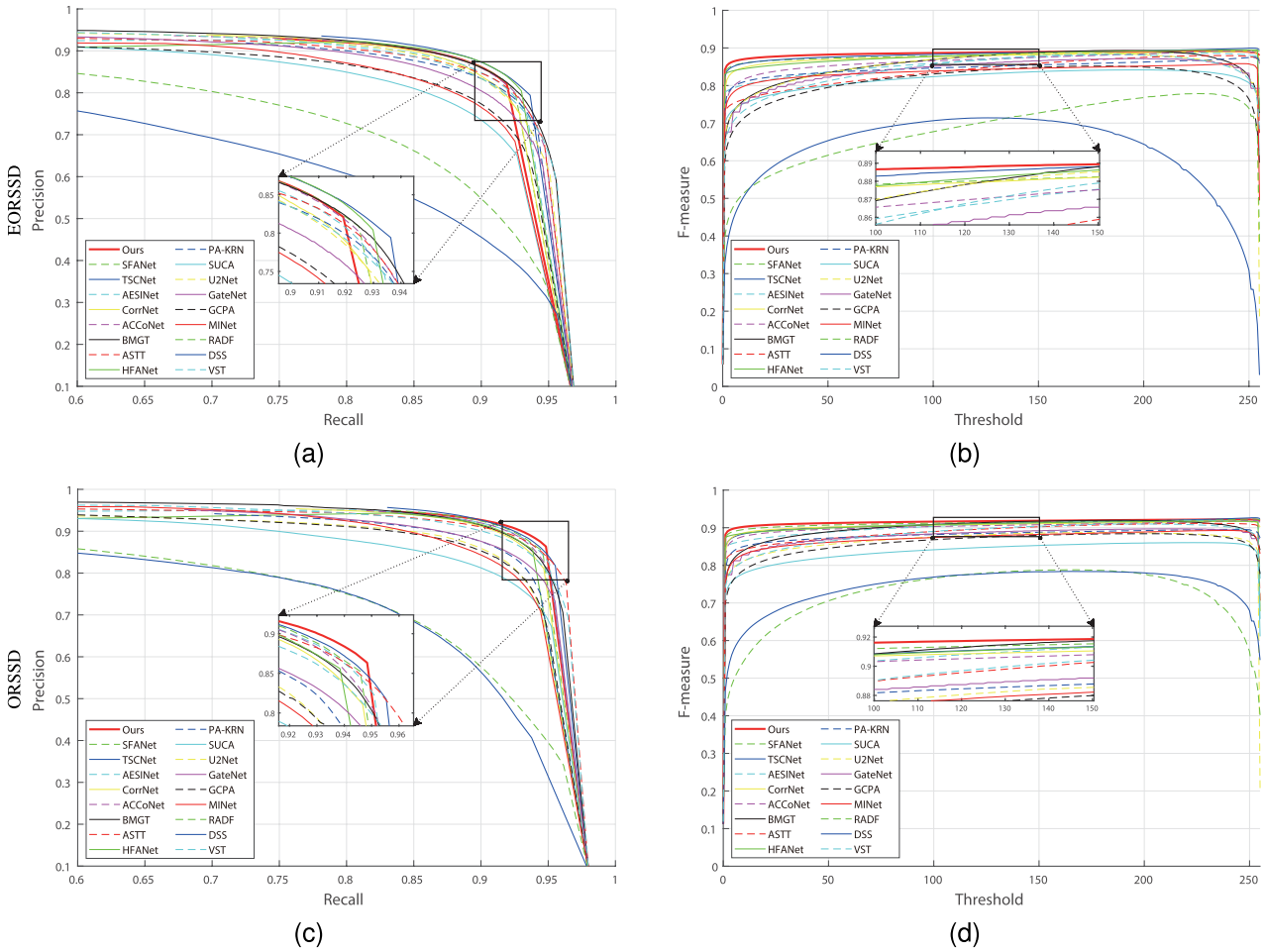


Fig. 5. (a) and (c) P–R curve and (b) and (d) F-measure curve of different methods on the testing dataset of EORSSD [12] and ORSSD [11]. Please zoom in for the best view of color.

(i.e., RRWR [55], DSG [56], and VST [32]), 13 CNN-based NSI-SOD methods (i.e., DSS [57], RADF [58], EGNet [9], PoolNet [59], CPDNet [60], MINet [8], GCPA [61], GateNet [62], U2Net [7], F3Net [63], SUCA [64], PA-KRN [10], and PGNet [65]), three traditional RSI-SOD methods (i.e., VOS [66], CMC [67], SMFF [68]), 16 CNN-based RSI-SOD methods (i.e., LVNet [11], SARNet [69], DAFNet [12], EMFINet [13], ERPNet [14], MSCNet [70], AGNet [28], FSMINet [29], HFANet [35], MJRBM [24], ACCoNet [54], SeaNet [16], CorrNet [15], PCDNet [71], ADSTNet [31], and TSCNet [72]), three graph-based RSI-SOD methods (i.e., RRNet [20], AESINet [30], and SFANet [22]), and two Transformer-based RSI-SOD methods (i.e., ASTT [23] and BMGT [36]). To ensure a fair comparison, all experiments were conducted using the same parameters on ORSSD, EORSSD, and ORSI-4199. Moreover, the data and saliency prediction maps of the respective models were obtained from publicly available sources or reimplemented from the provided source code. The comparison of quantitative and qualitative results is presented as follows.

1) Quantitative Comparison: Table I presents a quantitative comparison of our proposed method against other classical models, evaluated using S_α, F_β, E_ξ, and $\mathcal{M}$ on the EORSSD and ORSSD. It is evident that CNN-based NSI-SOD methods outperform traditional NSI-SOD methods. However, due to the unique characteristics of optical RSI datasets, there is still room for improvement when compared to RSI-SOD methods, particularly CNN-based RSI-SOD methods. Our proposed framework achieves overall superior performance compared to CNN-based RSI-SOD methods. Specifically, it secured four first-place rankings on the EORSSD. Although these results are not overwhelmingly dominant, MRBINet achieved a clean sweep on the ORSSD, ranking first across all eight metrics. Some methods, such as TSCNet and SFANet, outperform MRBINet on the EORSSD. However, MRBINet compensates for this difference by outperforming in F_β^{adp} by 0.0117 and 0.0134, and in E_ξ^{adp} by 0.0041 and 0.006, respectively. Furthermore, on the EORSSD, although MRBINet is slightly inferior to DAFNet by 0.0095 in $E_\xi^{\max}$, it shows satisfactory margins in other metrics, notably leading by 0.2199 in F_β^{adp} and also achieving 0.0923 and 0.043 in F_β^{mean} and E_ξ^{mean}, respectively. On the ORSSD, our method is the only one with $\mathcal{M}$ lower than 0.007, i.e., 0.0069. Additionally, our method demonstrates advantages over the second-ranking method TSCNet in S_α (0.9474 versus 0.9428) and F_β^{mean} (0.9110 versus 0.9030). Fig. 5 further illustrates the differences in P–R curves and F-measure curves. It is observed that while our model shows slight gaps in the P–R curve on the EORSSD compared to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

TSCNet and HFANet, which aligns with the results shown in Table I, MRBINet confidently outperforms other models on the remaining three visualized curves. These results substantiate that our model surpasses previous SOTA models.

The quantitative comparison results of multiple models on the challenging ORSI-4199 are presented in Table II, once again affirming the generalization capability of our model, given the significant size compared to EORSSD and ORSSD. Overall, our model achieved four first-place and four second-place rankings, decisively establishing its leading position. Compared to the second-ranked ASTT, our model lags behind in S_α, $E_\xi^{\max}$, and E_ξ^{mean}, but the largest difference is only 0.0023. However, we lead this model by 0.0228, which is a difference exceeding tenfold. Although the third-ranked TSCNet takes the lead on E_ξ^{adp}, our proposed method outperforms others across remaining metrics, the most significant difference lies in F_β^{adp}, achieving a margin of 0.0121. This indirectly validates that our model surpasses previous RSI-SOD models, and MRBINet demonstrates robustness and generalization capability.

Additionally, to enrich the model comparison results, we also measured model complexity and computational efficiency using three metrics: inference speed (Speed), network parameters (Params), and FLOPs, which were captured from publicly available RSI-SOD benchmarks [54], and our retraining. These data can be found in Table I. Encouragingly, our model's parameters and *FLOPs* are at a moderately high level in the overall comparison, for example, Params: 32.4M (Ours) versus 44.6M (ASTT) versus 68M (RRNet) and FLOPs: 42.8G (Ours) versus 69.45M (ASTT) versus 190M (RRNet). This reflects that our model is leading overall in the use of Transformer and graph methods. However, the Speed of MRBINet is only 9 fps, which falls short of real-time detection. In the next step, we will focus on investigating this issue further. Furthermore, for a more vivid demonstration of F_β^{adp}, FLOPs, and Params in multidimensional comparison, we used Fig. 6 to display the comparison results of numerous classical methods. It can be observed that although the FLOPs and Params of MRBINet are not among the top performers individually, MRBINet can exhibit competitive performance when considering all three metrics collectively.

2) Qualitative Comparison: To qualitatively demonstrate the competitiveness between the proposed model and SOTA methods, we selected several challenging scenarios for comparison, as shown in Fig. 7. These challenging scenarios include large-scale urban structures, tiny salient objects, objects with topological structures, low-light conditions, and regions with difficult-to-distinguish areas between or within salient objects. In the first scenario, although the salient objects are large, the cluttered lines inside and the complex background around them greatly interfere with the detection process. Compared to SFANet and RRNet, MRBINet can better explore the internal connections of the objects, accurately representing the salient objects, as seen in the first and third instances. In the second scenario, multiple tiny objects are present in the extensive background of the optical RSIs, with extremely small salient objects hidden among larger ones, posing a significant challenge to detection, as seen in the first instance. However, MRBINet successfully detects them, closely resembling the expected real-world performance. Conversely, ERPNet and AESINet exhibit instances of sticking or inaccurate representation. The difficulty of detecting topological objects is demonstrated in the third scenario, especially when the boundary features of the objects are complex. However, our model overcomes this challenge, demonstrating strong capabilities in depicting boundary features and determining overall object features. Although MJRBM and EGNet also enhance texture information using various methods, their efficacy does not significantly improve detection, resulting in information loss. In scenarios where the object color is similar to the background color and low-light contrast, models may experience errors in position localization and boundary locking, especially in dimly lit images, as seen in the first instance of the fourth scenario. Most models fail to delineate the riverbanks, while MRBINet predicts saliency maps more accurately, with clearer boundary features. In addition to the aforementioned scenarios, another challenging scenario involving densely packed microgaps is common in real-world scenarios. Small gaps between object objects or tiny gaps within objects' complex structures are crucial for precise object identification and classification. MRBINet accurately represents these microgaps in their true state, while SFANet, AESINet, and RRNet fail to successfully accomplish this task, resulting in information sticking. Overall, MRBINet can overcome various challenges across multiple scenarios and achieve relatively satisfactory results.

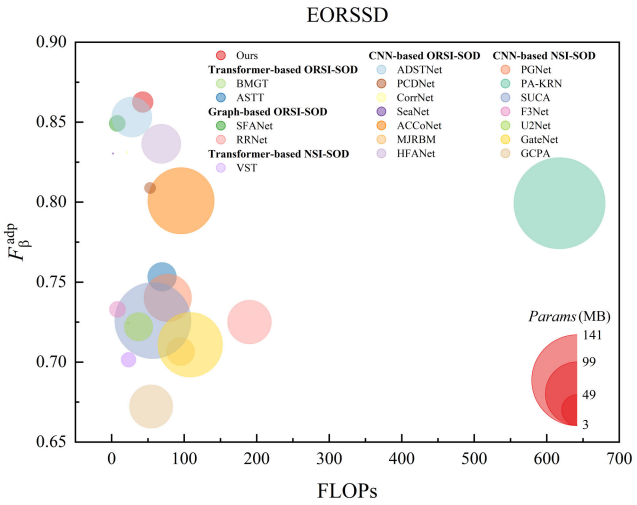


Fig. 6. Tradeoff comparisons of performance (F_β^{adp}, FLOPs, and Params) on the testing dataset. The bubble size is proportional to the parameters of the model.

C. Ablation Studies and Related Discussion

In this section, we conduct a series of ablation studies to validate the necessity of each module and configuration and explore the optimal weight values for the proposed MRBINet loss functions. All processes use the method of controlling variables, with each experiment modifying one parameter or component, consistent with the settings outlined in Section IV-A.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

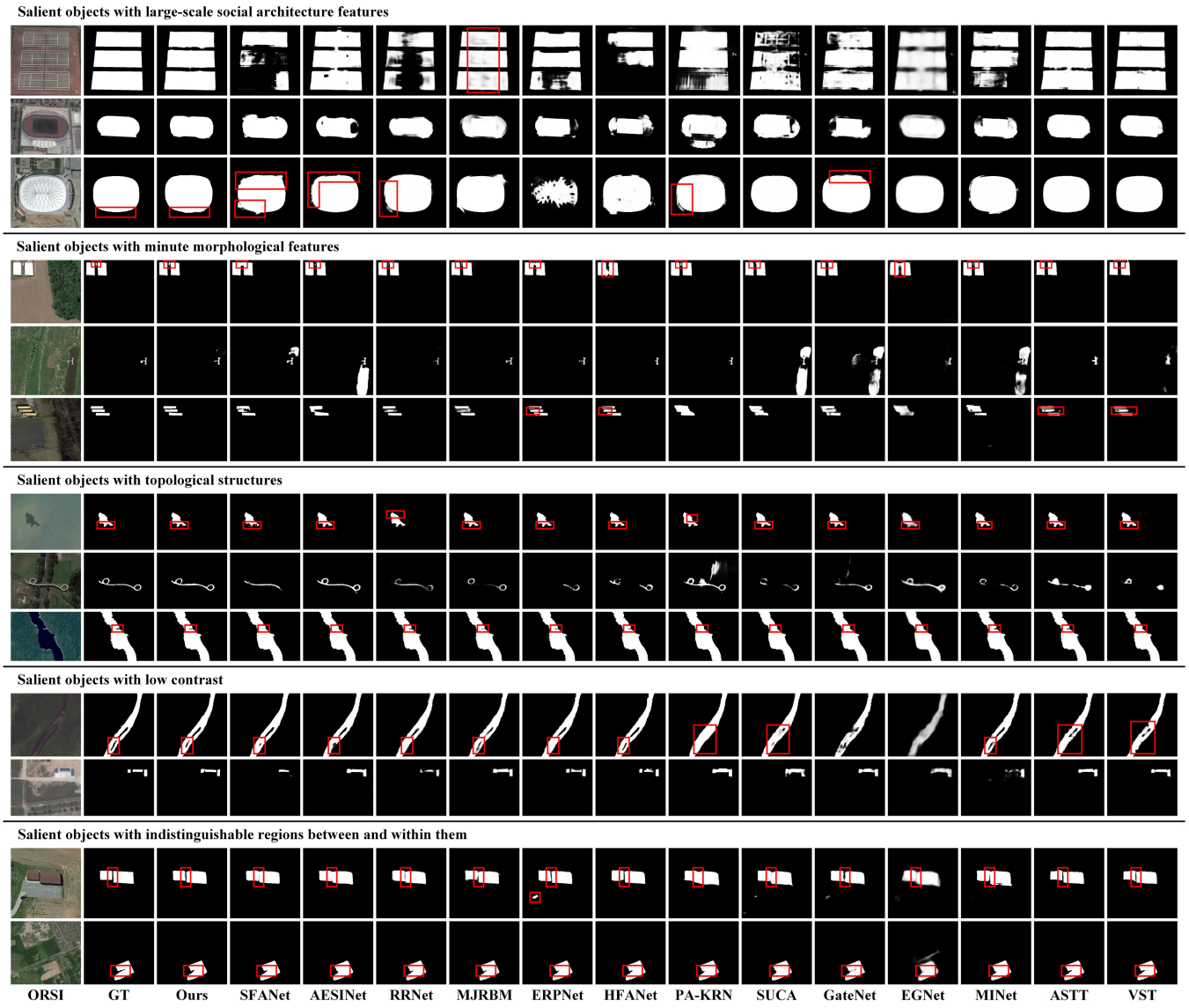


Fig. 7. Visual comparison of different saliency models, including one traditional NSI-SOD method (VST), five CNN-based NSI-SOD methods (MINet, EGNet, GateNet, SUCA, and PA-KRN), four CNN-based RSI-SOD methods (HFANet, ERPNet, MJRBM, and AESINet), two graph-based RSI-SOD methods (RRNet and SFANet), and one Transformer-based RSI-SOD method (ASTT).

1) Effectiveness of the Individual Modules: To validate the effectiveness of the proposed model, we explored its synergy through various combinations. First, we adopted Res2Net as the backbone, with each detection head configured with 3×3 convolutional kernels followed by 1×1 convolutions, serving as the baseline. Considering that the proposed model is based on boundary and region features, we primarily conducted subsequent component ablation experiments based on the BRS, as shown in Table III. Analysis of the table reveals a continuous improvement in network performance with the introduction of BGRR, MDC, and CECD. For instance, the introduction of BGRR, exploring the internal connection between boundary and region features, resulted in an improvement of S_α by 0.0021 and 0.0014 on two datasets compared to the baseline, respectively. Moreover, on ORSSD, the introduction of MDC and CECD led to improvements of 0.0072 and 0.0059 on $E_\xi^{\max}$, respectively.

TABLE III

ABLATION STUDY ON EVALUATING THE INDIVIDUAL CONTRIBUTION OF EACH CONTENT IN MRBINet. THE BEST RESULT IS INDICATED IN BOLD

No.	Baseline	BRS	BGRR	MDC	CECD	EORSSD [12]				ORSSD [11]			
						$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
1	✓					.9218	.8646	.9667	.0074	.9277	.8981	.9724	.0110
2	✓	✓	✓			.9239	.8650	.9679	.0070	.9291	.8996	.9771	.0102
3	✓	✓		✓		.9251	.8809	.9740	.0067	.9345	.9037	.9796	.0089
4	✓	✓			✓	.9253	.8805	.9731	.0069	.9324	.9018	.9783	.0097
5	✓	✓	✓	✓		.9316	.8827	.9762	.0061	.9366	.9054	.9807	.0082
6	✓	✓	✓		✓	.9327	.8854	.9744	.0060	.9451	.9129	.9835	.0074
7	✓	✓		✓	✓	.9269	.8842	**.9783**	.0059	.9372	.9059	.9822	.0079
8	✓	✓	✓	✓	✓	**.9351**	**.8852**	.9766	**.0056**	**.9474**	**.9199**	**.9851**	**.0069**

When BGRR, MDC, and CECD were jointly optimized in pairs and compared with MRBINet equipped with all three components, varying degrees of reduction were observed. For example, on EORSSD, the performance dropped to $\mathcal{M}$: 0.0056 (No.8) versus 0.0059 (No.7) versus 0.0060 (No.6) versus 0.0061 (No.5).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

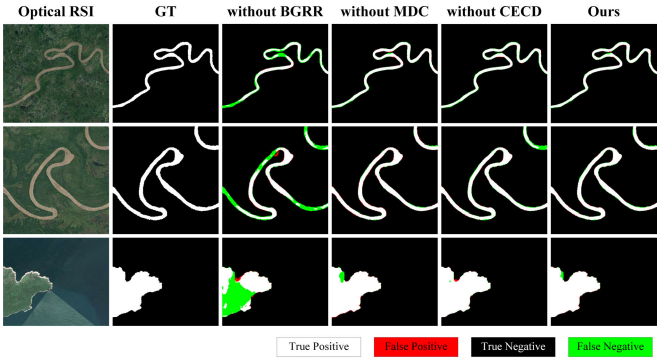


Fig. 8. Visualization of the contributions of BGRR, MDC, and CECD and the whole network.

Additionally, we presented significant prediction maps after removing BGRR, MDC, and CECD, as shown in Fig. 8, where different colors represent true positive (white), false positive (red), true negative (black), and false negative (green) results. The prediction map incorporating all components more closely resembles the GTs, while the absence of BGRR results in reduced recognition of internal object features, indirectly demonstrating BGRR's role in capturing internal relationships. Additionally, without BGRR, the region information becomes significantly diffuse, and without the constraint of boundary information, the generated prediction map exhibits varying degrees of false detections. Although the absence of MDC and CECD can still represent salient objects, there are varying degrees of information omission or extraneous filling along object edges, particularly when the CECD for final feature integration and refinement is missing. This also indirectly reflects that regions and boundaries are complementary sources of critical information. Overall, each module designed in this work makes an indispensable and crucial contribution to improving network performance, and the synergistic effect of each component helps elevate network performance to higher levels.

2) Ablation on GNN and Transformer: To further quantify the individual contributions of the GNN and Transformer modules in our architecture, we performed a targeted ablation study by isolating each component according to its insertion point. In our design, the GNN resides within the BGRR, while Transformer layers appear in both the MDC and CECD. For brevity, we refer to these three configurations as GNN in BGRR (GiB), Transformer in MDC (TiM), and Transformer in CECD (TiC). We then evaluated various combinations of GiB, TiM, and TiC to assess their impact on overall network performance (as shown in Table IV). The results clearly demonstrate that synergistic pairings of these modules consistently outperform any single-module variant. In particular, the configuration that applies the GNN in BGRR and the Transformer only in CECD achieves the second-best overall performance, ranking just behind the full model equipped with all components: on EORSSD, it matches the $F_\beta^{\max}$ of the complete network, and on ORSSD, it falls short in $E_\xi^{\max}$ by only 0.004. These findings confirm that both the GNN and Transformer are vital for representing salient features, and that their effective combination yields substantial cooperative gains in detection performance.

TABLE IV

ABLATION STUDY ON EVALUATING THE GNN AND TRANSFORMER OF BGRR, MDC, AND CECD. THE BEST RESULT IS INDICATED IN BOLD

No.	GiB	TiM	TiC	EORSSD [12]				ORSSD [11]			
				$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
1	✓			.9243	.8662	.9691	.0071	.9313	.9017	.9779	.0100
2		✓		.9266	.8816	.9755	.0066	.9357	.9045	.9795	.0083
3			✓	.9260	.8816	.9738	.0068	.9335	.9029	.9782	.0095
4	✓	✓		.9321	.8832	.9758	.0061	.9369	.9059	.9813	.0081
5	✓		✓	.9336	**.8852**	.9759	.0060	.9465	.9177	.9847	.0072
6		✓	✓	.9278	.8850	**.9766**	.0058	.9388	.9074	.9829	.0075
7	✓	✓	✓	**.9351**	**.8852**	**.9766**	**.0056**	**.9474**	**.9199**	**.9851**	**.0069**

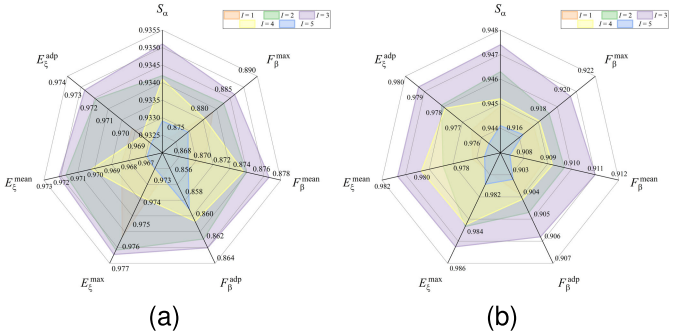


Fig. 9. Comparison of iteration times for erosion and dilation in BRS on (a) EORSSD [12] and (b) ORSSD [11].

3) Optimization of Iteration Times for Erosion and Dilation in BRS: We conducted experiments to analyze the optimal number of iterations for erosion and dilation operations within the BRS. The experimental results, depicted in Fig. 9, elucidate the impact of varying iteration numbers (I) on model performance. Specifically, at an iteration number of 1, the efficacy of erosion and dilation operations is limited, resulting in diminished performance. With increasing iterations to 2 and 3, these morphological operations effectively smooth and refine object regions, leading to a notable enhancement in performance. Notably, at an iteration number of 3, performance peaks, with the coverage area significantly surpassing that of other settings, indicative of an optimal balance between refining object regions and preserving essential features. However, further escalation of iteration numbers to 4 and beyond precipitates excessive morphological operations, resulting in loss of detail and unwarranted smoothing of object regions, thereby compromising performance. In particular, at an iteration number of 5, performance markedly declines, underscoring the adverse impact of excessive morphological processing on image fidelity and structural integrity. In conclusion, based on our experimental findings, we ascertain that the optimal iteration number for erosion and dilation operations in BRS is 3. This parameter configuration facilitates optimal performance while mitigating the negative repercussions of excessive iterations.

4) Visualization of Information Filtering in the SFE: To demonstrate the role of SFE in information filtering, we visualized the features to explore the contribution of this model to reducing noise interference. As shown in Fig. 10, we selected two instances with relatively complex backgrounds for this visualization. In the absence of SFE, the attention of MRBINet

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

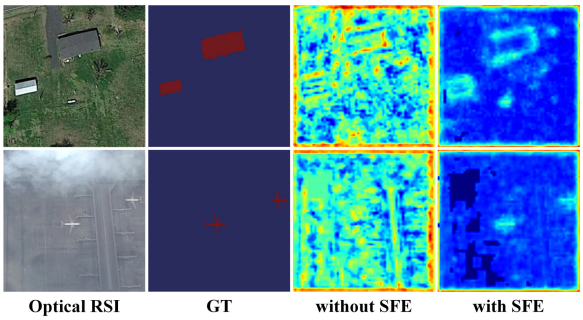


Fig. 10. Feature visualization to show the contribution of SFE.

TABLE V

PERFORMANCE EVALUATION ACROSS VARIOUS ENCODER BACKBONES IN MRBINet. THE BEST RESULT IN EACH COLUMN IS BOLD

Method	EORSSD [12]				ORSSD [11]			
	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
MRBINet-VGG	.9291	.8694	.9668	.0070	.9353	.8936	.9692	.0107
MRBINet-ResNet	.9325	.8763	.9702	.0059	.9438	.9039	.9778	.0079
MRBINet-Res2Net	**.9351**	**.8768**	**.9721**	**.0056**	**.9474**	**.9110**	**.9812**	**.0069**

is easily distracted by background elements, both in the frequency and spatial domains. In the spatial domain, background features often introduce noise that diverts focus away from the salient object, leading to less accurate feature extraction. In the frequency domain, irrelevant high-frequency components from the background can cause interference, further degrading the model's attention to the object. However, after introducing SFE, the model is able to effectively isolate the object of interest by enhancing spatial focus and reducing unwanted high-frequency noise from the background. This allows the model to concentrate its attention more precisely on the object, resulting in a more accurate representation of its features. Although some discrepancies remain when compared to the GT, the object's shape is now more clearly discernible. Overall, SFE improves the model's ability to detect salient objects by combining spatial-domain and frequency-domain processing.

5) Flexibility of the Proposed Approach: To validate the flexibility of the proposed MRBINet, we designed two additional variants based on the classic backbone VGG [73] and ResNet [74], denoted as MRBINet-VGG and MRBINet-ResNet, as shown in Table V. These variants were extensively evaluated on EORSSD and ORSSD. As shown in Table V, compared to using Res2Net as the backbone (MRBINet-Res2Net), both new variants exhibit considerable room for improvement, for example, on the EORSSD, E_ξ^{mean}: 0.9721 (MRBINet-Res2Net) versus 0.9702 (MRBINet-ResNet) versus 0.9668 (MRBINet-VGG). Additionally, MRBINet-VGG performs worse than MRBINet-ResNet, ranking lowest among the three variants. This indirectly confirms that the primary features extracted by the backbone significantly contribute to the overall network performance. The stronger the extraction capability of the backbone, the richer the features it provides. In conclusion, MRBINet demonstrates substantial adaptability to multiple backbones and effectively utilizes the encoded features from these networks to their full potential.

TABLE VI

ABLATION STUDY ON THE EFFECTIVENESS OF $\ell_{\text{BCE}}^{\text{S}}$, $\ell_{\text{IoU}}^{\text{S}}$, $\ell_{\text{FLoss}}^{\text{S}}$, AND $\ell_{\text{BCE}}^{\text{B}}$. THE BEST RESULT IN EACH COLUMN IS BOLD

No.	ℓ_{BCE}^{S}	ℓ_{IoU}^{S}	ℓ_{Flodss}^{S}	ℓ_{BCE}^{B}	EORSSD [12]				ORSSD [11]			
					$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
1	✓				.9170	.8611	.9514	.0077	.9313	.9057	.9617	.0095
2		✓			.9142	.8581	.9507	.0085	.9158	.8922	.9519	.0112
3			✓		.9185	.8627	.9518	.0076	.9333	.9097	.9643	.0091
4	✓	✓			.9219	.8666	.9558	.0065	.9344	.9104	.9773	.0083
5	✓		✓		.9231	.8672	.9567	.0064	.9434	.9129	.9793	.0075
6		✓	✓		.9239	.8773	.9618	.0062	.9400	.9116	.9785	.0078
7	✓	✓	✓		.9247	.8795	.9636	.0059	.9445	.9138	.9806	.0075
8	✓	✓	✓	✓	**.9251**	**.8810**	**.9653**	**.0057**	**.9452**	**.9146**	**.9823**	**.0073**

Through feature fusion and refinement, it enhances the effectiveness of RSI-SOD.

6) Complementarity of loss function: To further explore the auxiliary roles of each loss function within the MRBINet, we examined the effects of loss functions $\mathcal{L}_{\text{BCE}}^{\text{S}}$, $\mathcal{L}_{\text{IoU}}^{\text{S}}$, and $\mathcal{L}_{\text{Floss}}^{\text{S}}$ for region features and $\mathcal{L}_{\text{BCE}}^{\text{B}}$ for boundary features. Here, we set the weights of the above variables to 1. Since the boundary features are introduced as an additional component to optimize performance further, we first focused on the study of $\mathcal{L}_{\text{BCE}}^{\text{S}}$, $\mathcal{L}_{\text{IoU}}^{\text{S}}$, and $\mathcal{L}_{\text{Floss}}^{\text{S}}$ for regions, as shown in the first seven rows of Table VI. It is evident that when only $\mathcal{L}_{\text{IoU}}^{\text{S}}$ is used, the performance is at the lowest overall level. However, with the progressive introduction of $\mathcal{L}_{\text{BCE}}^{\text{S}}$ and $\mathcal{L}_{\text{Floss}}^{\text{S}}$, significant performance improvements are observed. The most considerable improvement is seen on the ORSSD, where $E_\xi^{\max}$ increases by 0.0266, from 0.9519 (No.2) to 0.9785 (No.7). The combination $\mathcal{L}_{\text{BCE}}^{\text{S}}$, $\mathcal{L}_{\text{IoU}}^{\text{S}}$, and $\mathcal{L}_{\text{Floss}}^{\text{S}}$ achieves the optimal performance compared to any single or partial combination. Moreover, we investigated the scenario of additionally incorporating $\mathcal{L}_{\text{BCE}}^{\text{B}}$. The results show that this configuration achieved the best performance across all comparison data, demonstrating the indispensable contribution of all loss functions to maximizing network performance.

Furthermore, we conducted extensive experiments to determine the optimal loss function weight values for region features. Given that $\mathcal{L}_{\text{BCE}}$ primarily handles the classification of object regions and backgrounds, its contribution to region features is relatively stable. Using this as a baseline, we explored the optimal weight values for $\mathcal{L}_{\text{IoU}}^{\text{S}}$ and $\mathcal{L}_{\text{Floss}}^{\text{S}}$, denoted as γ and ψ, respectively, as illustrated in Fig. 11. Initially, we set ψ to 0 to conduct experiments. Observations indicate that as γ increases, the network's performance on EORSSD and ORSSD consistently improves, reaching optimal performance when γ equals 0.6. Subsequently, using γ equals 0.6 as the foundation, we continued to explore the optimal value for ψ. Similar to the search for the optimal γ value, as the ψ parameter increases, network performance improves. When ψ reaches 0.9, MRBINet performance peaks; beyond this point, specifically at ψ is 1, performance begins to decline.

In summary, the designed combination of loss functions is effective, and MRBINet achieves its maximum potential when γ and ψ are set to 0.6 and 0.9, respectively.

D. Analysis of Failure Samples

We conducted an in-depth investigation into some failure cases of our proposed MRBINet to gather insights for future

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

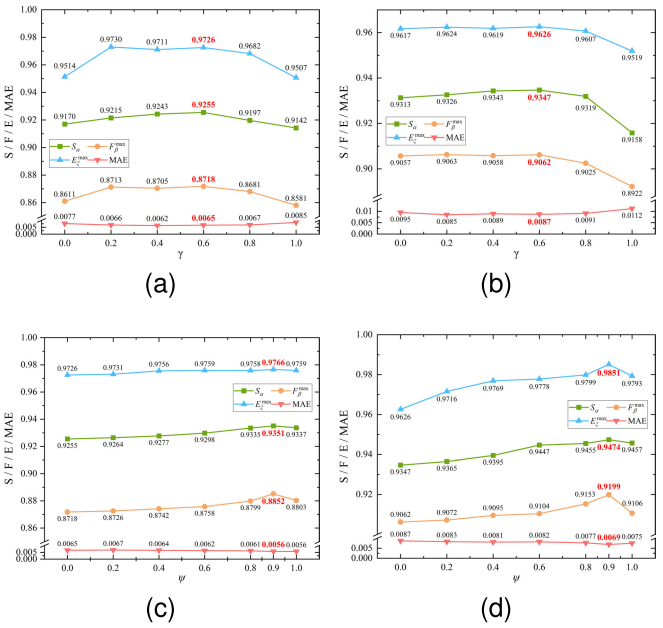


Fig. 11. Performance trends are produced with different values of the hyperparameter γ and ψ. The curves of S_α, $F_\beta^{\max}$, $E_\xi^{\max}$, and $\mathcal{M}$ on (a) and (c) EORSSD [12] and (b) and (d) ORSSD [11].

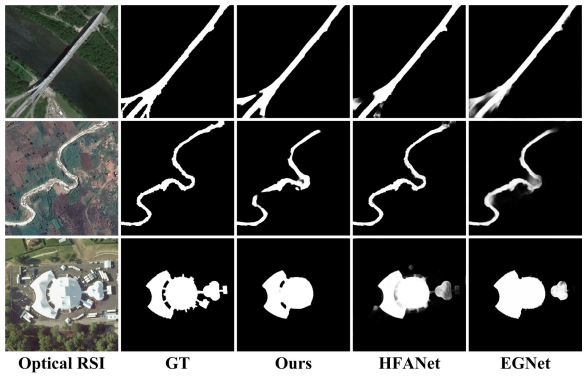


Fig. 12. Visualization results depict the failure cases encountered on the ORSI-4199 [24], particularly with challenging scenes.

model optimizations. In Fig. 12, we present failure examples of MRBINet in the challenging scenarios of the ORSI-4199, where HFANet and EGNet also exhibited similar errors. Specifically, although the relationship between boundary and region features has been extensively explored, significant interference occurs when complex structures are located at the image boundaries, and there are multilayered color gradations. This interference leads to blurred object detection, though MRBINet performs closer to the GT, as seen in the first example. In the second and third examples, the object exhibits significant internal color variation and complex internal structures. Visible external interferences were not considered part of the salient object, leading to errors in many models, and causing loss of information or blurred regions influenced by the background. Moving forward, we will continue to analyze the inherent complexities of optical RSI datasets and incorporate cognitive analysis methods into network design to achieve greater efficacy.

V. Conclusion

In this work, we propose MRBINet to address challenges in RSI-SOD, utilizing a set of strategies that enable effective interaction between boundary and region features through several specialized modules. Specifically, the BRS extracts boundary and region features from images, while the BGRR uncovers complementary semantic relationships between these features, facilitating efficient modeling of topological structures. The MDC captures complex relationships between the interiors of regions and their boundaries, yielding more complete and coherent salient objects in the detection map. Finally, the CECD employs a multistage cascading approach to integrate boundary and region features, thereby reinforcing critical object characteristics. Through these modules, region information effectively fills boundary interiors, while boundary features act as constraints, limiting the freedom of object information to enhance coherence across the detection map. Extensive evaluations on three public RSI-SOD benchmark datasets demonstrate that MRBINet shows competitive performance compared to SOTA methods. Various visual comparisons further illustrate the effectiveness of each component. Future work will focus on the characteristics of optical RSIs datasets, combining refined boundaries and chromatic differentiation methods, to accurately separate salient objects in complex backgrounds with more lightweight techniques while ensuring detection accuracy.

References

[1] Y. Shi, S. Zhao, J. Wu, Z. Wu, and H. Yan, "Fixated object detection based on saliency prior in traffic scenes," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 3, pp. 1413–1426, Mar. 2024.

[2] Y. Jia, J. Zhao, and L. Yu, "AADH-YOLOv5: Improved YOLOv5 based on adaptive activate decoupled head for garbage detection," *J. Electron. Imag.*, vol. 32, no. 4, Jul. 2023, Art. no. 043017.

[3] W. Liu, G. Lin, T. Zhang, and Z. Liu, "Guided co-segmentation network for fast video object segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 4, pp. 1607–1617, Apr. 2021.

[4] C. Chen et al., "TOPIQ: A top-down approach from semantics to distortions for image quality assessment," *IEEE Trans. Image Process.*, vol. 33, pp. 2404–2418, 2024.

[5] P. Gao, X.-Y. Zhang, X.-L. Yang, F. Gao, H. Fujita, and F. Wang, "Robust visual tracking with extreme point graph-guided annotation: Approach and experiment," *Expert Syst. Appl.*, vol. 238, Mar. 2024, Art. no. 122013.

[6] Y. Lin, D. Zhang, X. Fang, Y. Chen, K.-T. Cheng, and H. Chen, "Rethinking boundary detection in deep learning models for medical image segmentation," in *Proc. Int. Conf. Inf. Process. Med. Imag.* Cham, Switzerland: Springer, 2023, pp. 730–742.

[7] X. Qin, Z. Zhang, C. Huang, M. Dehghan, O. R. Zaiane, and M. Jagersand, "U2-Net: Going deeper with nested U-structure for salient object detection," *Pattern Recognit.*, vol. 106, Oct. 2020, Art. no. 107404.

[8] Y. Pang, X. Zhao, L. Zhang, and H. Lu, "Multi-scale interactive network for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9413–9422.

[9] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8779–8788.

[10] B. Xu, H. Liang, R. Liang, and P. Chen, "Locate globally, segment locally: A progressive architecture with knowledge review network for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 4, pp. 3004–3012.

[11] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

[12] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.

[13] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605315.

[14] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.

[15] G. Li, Z. Liu, Z. Bai, W. Lin, and H. Ling, "Lightweight salient object detection in optical remote sensing images via feature correlation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617712.

[16] G. Li, Z. Liu, X. Zhang, and W. Lin, "Lightweight salient object detection in optical remote-sensing images via semantic matching and edge alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5601111.

[17] P. Han, B. Zhao, and X. Li, "Progressive feature interleaved fusion network for remote-sensing image salient object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2023, Art. no. 5500414.

[18] Y. Zhang, J. Guo, H. Yue, X. Yin, and S. Zheng, "Transformer guidance dual-stream network for salient object detection in optical remote sensing images," *Neural Comput. Appl.*, vol. 35, no. 24, pp. 17733–17747, Aug. 2023.

[19] Z. Bai, G. Li, and Z. Liu, "Global–local–global context-aware network for salient object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 198, pp. 184–196, Apr. 2023.

[20] R. Cong, Y. Zhang, L. Fang, J. Li, Y. Zhao, and S. Kwong, "RRNet: Relational reasoning network with parallel multiscale attention for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613311.

[21] Y. Liu, Y. Yuan, and Q. Wang, "Uncertainty-aware graph reasoning with global collaborative learning for remote sensing salient object detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[22] Y. Quan, H. Xu, R. Wang, Q. Guan, and J. Zheng, "ORSI salient object detection via progressive semantic flow and uncertainty-aware refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5608013.

[23] L. Gao, B. Liu, P. Fu, and M. Xu, "Adaptive spatial tokenization transformer for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602915.

[24] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607913.

[25] H. Sun et al., "Defense against adversarial cloud attack on remote sensing salient object detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Aug. 2024, pp. 8345–8354.

[26] H. Qin, T. Xu, P. Liu, J. Xu, and J. Li, "DMSSN: Distilled mixed spectral–spatial network for hyperspectral salient object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512618.

[27] Y. Lian, X. Shi, S. Shen, and J. Hua, "Multitask learning for image translation and salient object detection from multimodal remote sensing images," *Vis. Comput.*, vol. 40, no. 3, pp. 1395–1414, Mar. 2024.

[28] Y. Lin, H. Sun, N. Liu, Y. Bian, J. Cen, and H. Zhou, "Attention guided network for salient object detection in optical remote sensing images," in *Proc. Int. Conf. Artif. Neural Netw.* Cham, Switzerland: Springer, 2022, pp. 25–36.

[29] K. Shen, X. Zhou, B. Wan, R. Shi, and J. Zhang, "Fully squeezed multiscale inference network for fast and accurate saliency detection in optical remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[30] X. Zeng, M. Xu, Y. Hu, H. Tang, Y. Hu, and L. Nie, "Adaptive edge-aware semantic interaction network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6507705.

[31] J. Zhao, Y. Jia, L. Ma, and L. Yu, "Adaptive dual-stream sparse transformer network for salient object detection in optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5173–5192, 2024.

[32] N. Liu, N. Zhang, K. Wan, L. Shao, and J. Han, "Visual saliency transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 4722–4732.

[33] Z. Wang, Y. Zhang, Y. Liu, Z. Wang, S. Coleman, and D. Kerr, "TF-SOD: A novel transformer framework for salient object detection," *Neural Comput. Appl.*, vol. 34, no. 14, pp. 11789–11806, Jul. 2022.

[34] Y. Ke Yun and W. Lin, "SelfReformer: Self-refined network with transformer for salient object detection," 2022, *arXiv:2205.11283*.

[35] Q. Wang, Y. Liu, Z. Xiong, and Y. Yuan, "Hybrid feature aligned network for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624915.

[36] K. Huang, C. Tian, and G. Li, "Bidirectional mutual guidance transformer for salient object detection in optical remote sensing images," *Int. J. Remote Sens.*, vol. 44, no. 13, pp. 4016–4033, Jul. 2023.

[37] G. Li, Z. Bai, Z. Liu, X. Zhang, and H. Ling, "Salient object detection in optical remote sensing images driven by transformer," *IEEE Trans. Image Process.*, vol. 32, pp. 5257–5269, 2023.

[38] M. Gori, G. Monfardini, and F. Scarselli, "A new model for learning in graph domains," in *Proc. IEEE Int. Joint Conf. Neural Netw.*, Mar. 2005, pp. 729–734.

[39] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Trans. Neural Netw.*, vol. 20, no. 1, pp. 61–80, Dec. 2008.

[40] A. Micheli, "Neural network for graphs: A contextual constructive approach," *IEEE Trans. Neural Netw.*, vol. 20, no. 3, pp. 498–511, Mar. 2009.

[41] A. Fout, J. Byrd, B. Shariat, and A. Ben-Hur, "Protein interface prediction using graph convolutional networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Nov. 2017, pp. 6530–6539.

[42] F. Monti, M. Bronstein, and X. Bresson, "Geometric matrix completion with recurrent multi-graph neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[43] R. Ying, R. He, K. Chen, P. Eksombatchai, W. L. Hamilton, and J. Leskovec, "Graph convolutional neural networks for web-scale recommender systems," in *Proc. 24th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Nov. 2018, pp. 974–983.

[44] T. Hamaguchi, H. Oiwa, M. Shimbo, and Y. Matsumoto, "Knowledge transfer for out-of-knowledge-base entities: A graph neural network approach," 2017, *arXiv:1706.05674*.

[45] S. Liang, Z. Hua, and J. Li, "GCN-based multi-scale dual fusion for remote sensing building change detection," *Int. J. Remote Sens.*, vol. 44, no. 3, pp. 953–980, Feb. 2023.

[46] W. Li, Q. Liu, S. Fan, H. Bai, and M. Xin, "Multistage superpixel-guided hyperspectral image classification with sparse graph attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5519718.

[47] F. Liu, Z. Hua, J. Li, and L. Fan, "MFBGR: Multi-scale feature boundary graph reasoning network for polyp segmentation," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106213.

[48] Y. Meng et al., "BI-GCN: Boundary-aware input-dependent graph convolution network for biomedical image segmentation," 2021, *arXiv:2110.14775*.

[49] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, and P. Torr, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2021.

[50] B. Jähne, *Digital Image Processing*. Cham, Switzerland: Springer, 2005.

[51] K. Wang, X. Zhang, Y. Lu, X. Zhang, and W. Zhang, "CGRNet: Contour-guided graph reasoning network for ambiguous biomedical image segmentation," *Biomed. Signal Process. Control*, vol. 75, May 2022, Art. no. 103621.

[52] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[53] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[54] G. Li, Z. Liu, D. Zeng, W. Lin, and H. Ling, "Adjacent context coordination network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 526–538, Jan. 2023.

[55] C. Li, Y. Yuan, W. Cai, Y. Xia, and D. D. Feng, "Robust saliency detection via regularized random walks ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 2710–2717.

[56] L. Zhou, Z. Yang, Z. Zhou, and D. Hu, "Salient region detection using diffusion process on a two-layer sparse graph," *IEEE Trans. Image Process.*, vol. 26, no. 12, pp. 5882–5894, Dec. 2017.

[57] Q. Hou, M.-M. Cheng, X. Hu, A. Borji, Z. Tu, and P. Torr, "Deeply supervised salient object detection with short connections," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3203–3212.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.

[58] X. Hu, L. Zhu, J. Qin, C.-W. Fu, and P.-A. Heng, "Recurrently aggregating deep features for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2018, vol. 32, no. 1, pp. 6943–6950.

[59] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 3917–3926.

[60] C. Xie, C. Xia, M. Ma, Z. Zhao, X. Chen, and J. Li, "Pyramid grafting network for one-stage high resolution saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11707–11716.

[61] Z. Chen, Q. Xu, and R. Cong, "Global context-aware progressive aggregation network for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 7, pp. 10599–10606.

[62] X. Zhao, Y. Pang, L. Zhang, H. Lu, and L. Zhang, "Suppress and balance: A simple gated network for salient object detection," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, Aug. 2020, pp. 35–51.

[63] J. Wei, S. Wang, and Q. Huang, "F^3Net: Fusion, feedback and focus for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2020, vol. 34, no. 7, pp. 12321–12328.

[64] J. Li, Z. Pan, Q. Liu, and Z. Wang, "Stacked U-shape network with channel-wise attention for salient object detection," *IEEE Trans. Multimedia*, vol. 23, pp. 1397–1409, 2020.

[65] Z. Wu, L. Su, and Q. Huang, "Cascaded partial decoder for fast and accurate salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3907–3916.

[66] Q. Zhang, L. Zhang, W. Shi, and Y. Liu, "Airport extraction via complementary saliency analysis and saliency-oriented active contour model," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 7, pp. 1085–1089, Jul. 2018.

[67] Z. Liu, D. Zhao, Z. Shi, and Z. Jiang, "Unsupervised saliency model with color Markov chain for oil tank detection," *Remote Sens.*, vol. 11, no. 9, p. 1089, May 2019.

[68] L. Zhang, Y. Liu, and J. Zhang, "Saliency detection based on self-adaptive multiple feature fusion for remote sensing images," *Int. J. Remote Sens.*, vol. 40, no. 22, pp. 8270–8297, Nov. 2019.

[69] Z. Huang, H. Chen, B. Liu, and Z. Wang, "Semantic-guided attention refinement network for salient object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 11, p. 2163, May 2021.

[70] Y. Lin, H. Sun, N. Liu, Y. Bian, J. Cen, and H. Zhou, "A lightweight multi-scale context network for salient object detection in optical remote sensing images," in *Proc. 26th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2022, pp. 238–244.

[71] K. Huang, C. Tian, and C.-W. Lin, "Progressive context-aware dynamic network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4703816.

[72] G. Li, Z. Bai, and Z. Liu, "Texture-semantic collaboration network for ORSI salient object detection," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 4, pp. 2464–2468, Apr. 2024.

[73] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[74] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.



Yun Jia (Member, IEEE) received the B.E., M.E., and Ph.D. degrees in information and communication engineering from the College of Information and Communication Engineering, Harbin Institute of Technology, Harbin, China, in 2006, 2009, and 2014, respectively.

He was a Visiting Scholar with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 2011 to 2012. He is currently with Shandong Technology and Business University, Yantai, China. His research interests include cognitive radio/software radio, deep learning, and intelligent signal processing.



Jie Zhao received the B.E. degree in computer science and technology from Shandong Youth University of Political Science, Jinan, China, in 2021. He is currently pursuing the M.E. degree in computer science and technology with the School of Information and Electronic Engineering, Shandong Technology and Business University, Yantai, Shandong, China.

His research interests include remote sensing image interpretation, saliency detection and segmentation, and deep learning.



Lin Ma (Senior Member, IEEE) received the B.E., M.E., and Ph.D. degrees in information and communication engineering from the College of Information and Communication Engineering, Harbin Institute of Technology, Harbin, China, in 2003, 2005, and 2009, respectively.

From 2013 to 2014, he was a Visiting Scholar at the University of Toronto, Toronto, ON, Canada. He is currently a Professor with Harbin Institute of Technology. His research interests include image processing, artificial intelligence-based cognition and recognition technology for complex environments, and deep learning.



Lidan Yu received the B.E. degree in computer science and technology from Qingdao University of Science and Technology, Qingdao, China, in 2021, and the M.E. degree in computer science and technology from the School of Information and Electronic Engineering, Shandong Technology and Business University, Shandong, China, in 2024.

She is currently a Lecturer with the School of Computer and Information Engineering, Qilu Institute of Technology, Jinan, China. Her research interests include image processing and computer vision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:08:35 UTC from IEEE Xplore. Restrictions apply.